

BFS Traversal

Step 1: Start at E

Legend

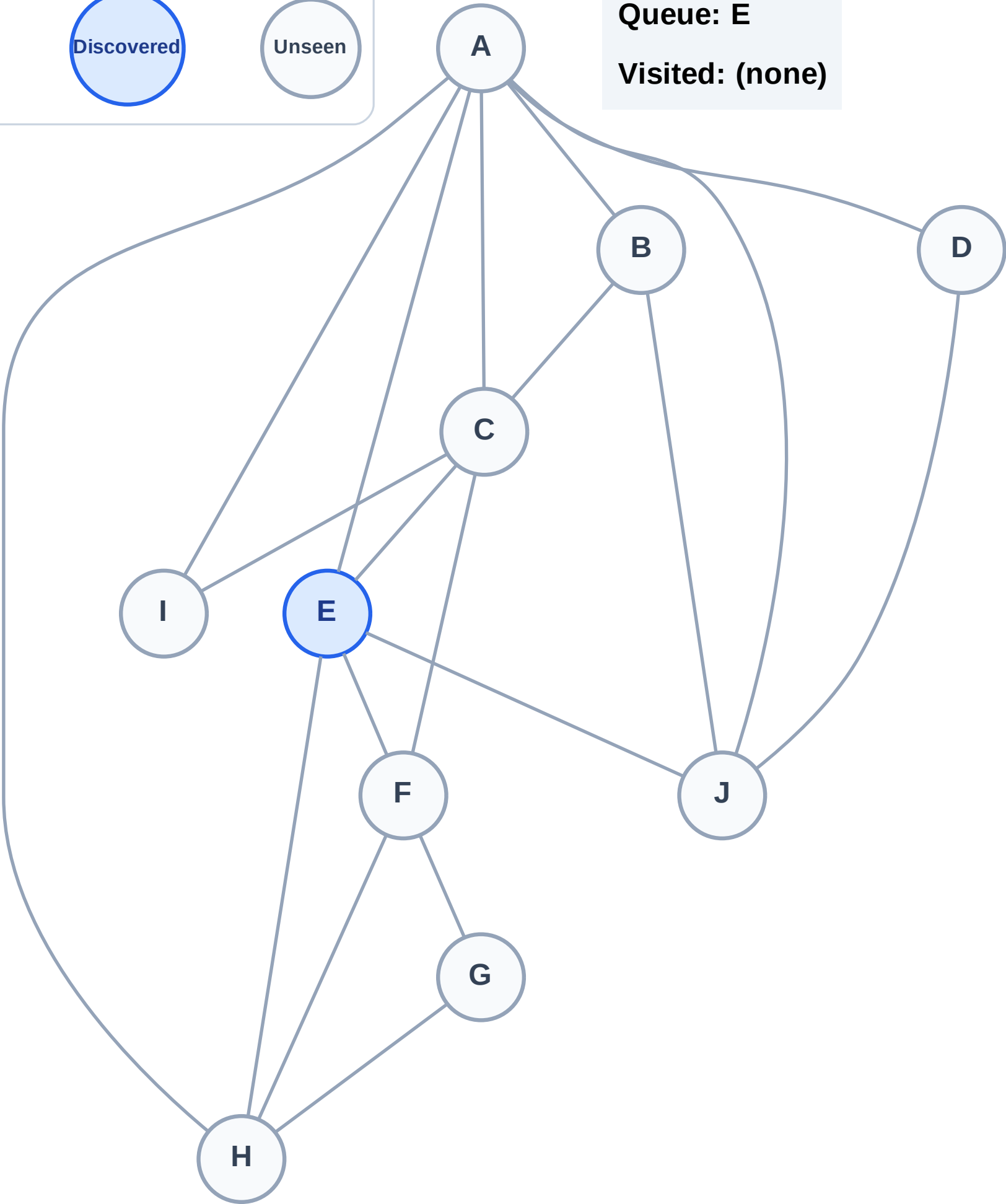
Complete

Processing

Discovered

Unseen

Queue: E
Visited: (none)



BFS Traversal

Step 2: Process node E

Legend

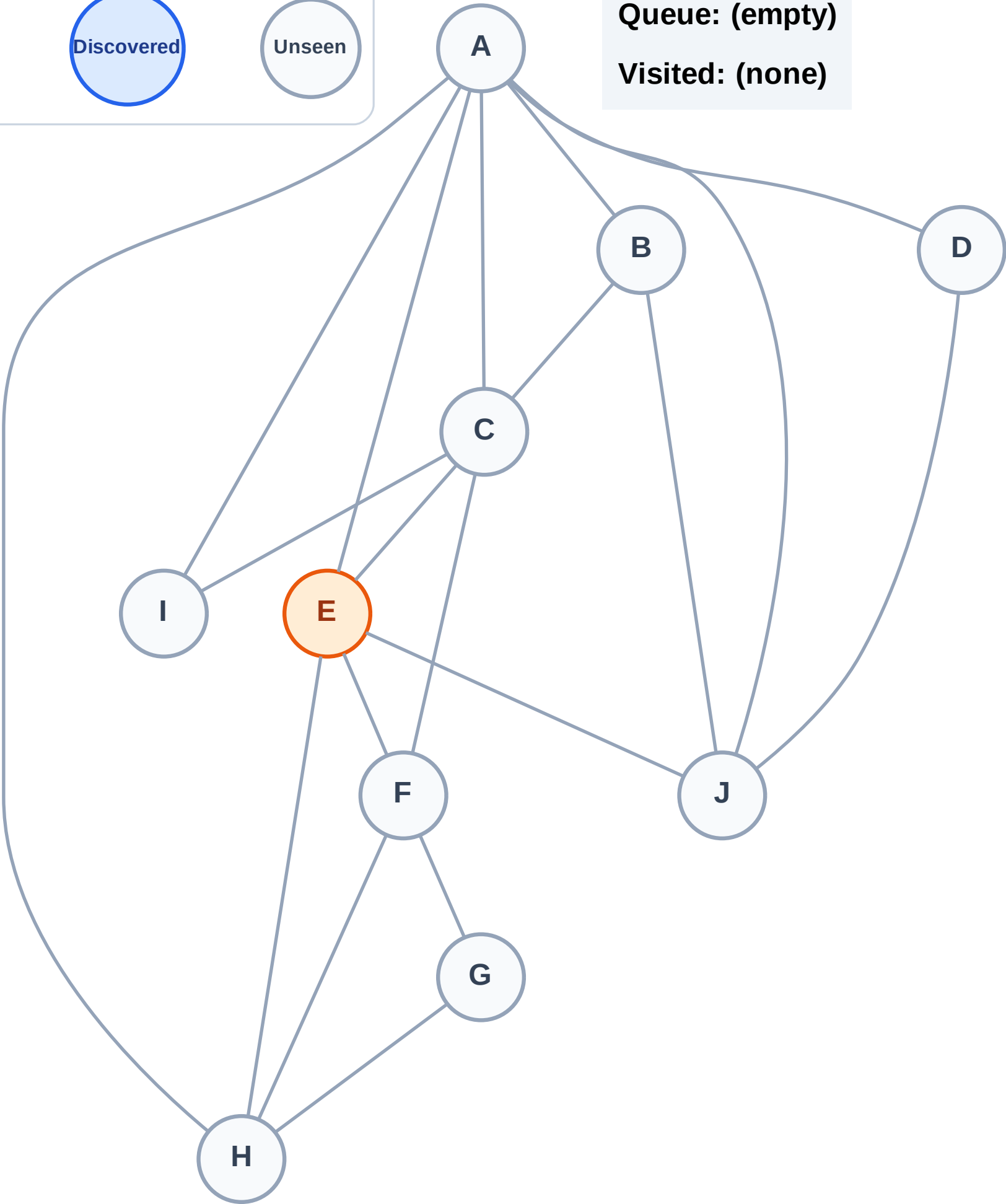
Complete

Processing

Discovered

Unseen

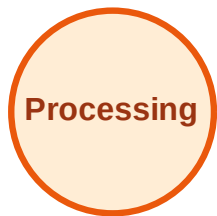
Queue: (empty)
Visited: (none)



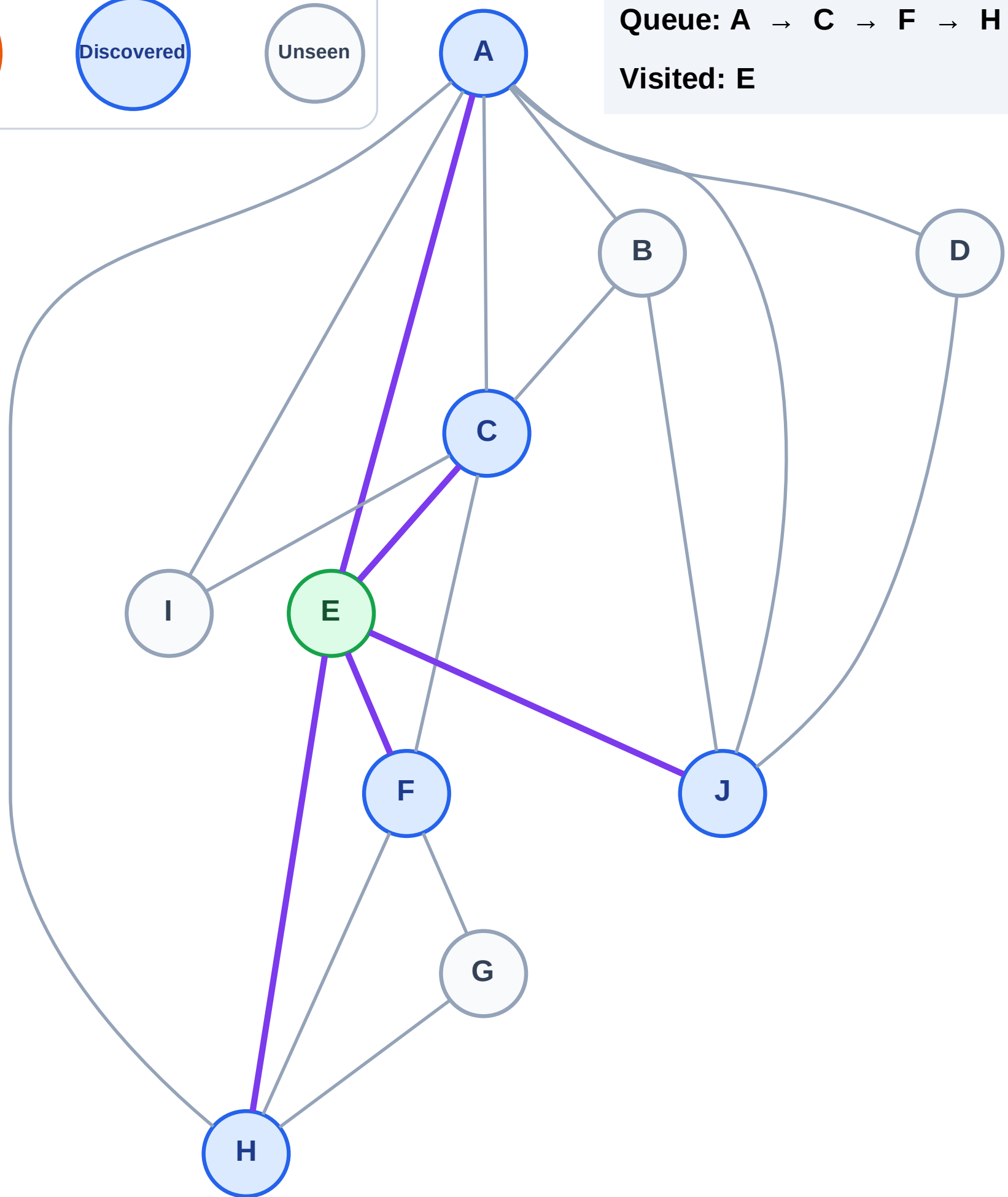
BFS Traversal

Step 3: Discover A, C, F, H, J from E

Legend



Queue: A → C → F → H → J
Visited: E



BFS Traversal

Step 4: Process node A

Legend

Complete

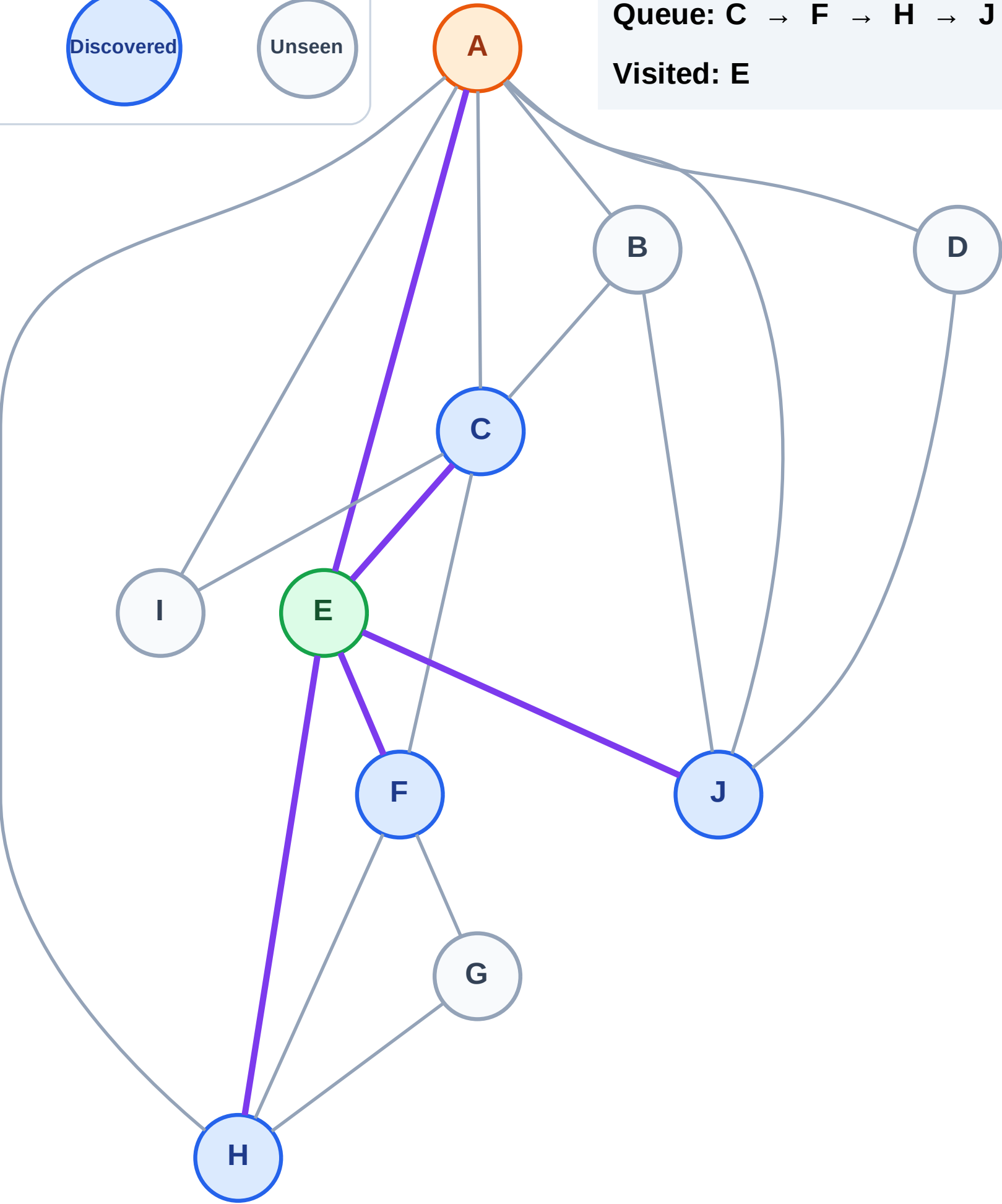
Processing

Discovered

Unseen

Queue: C → F → H → J

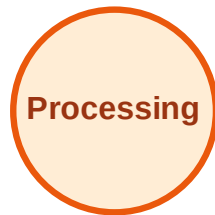
Visited: E



BFS Traversal

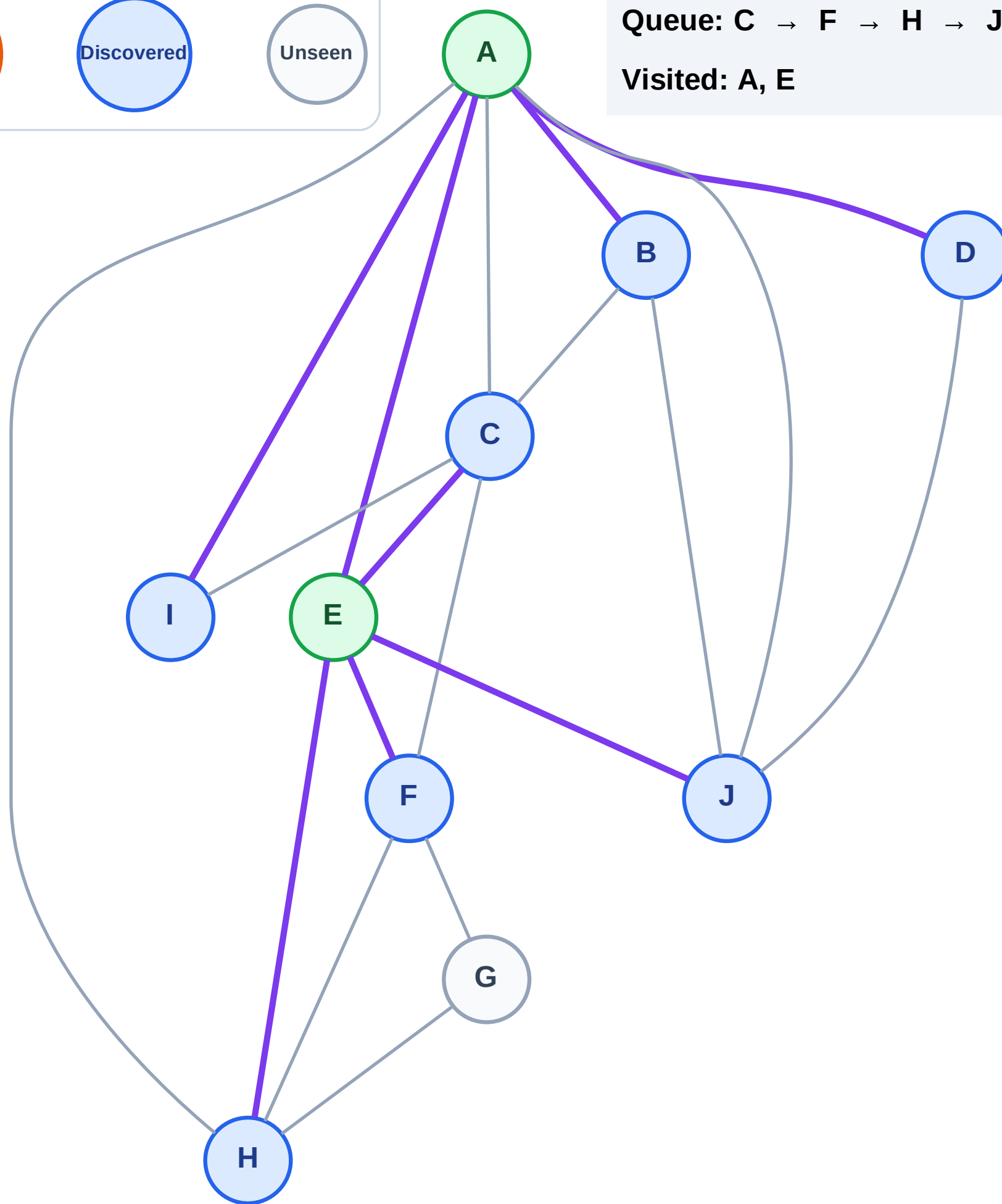
Step 5: Discover B, D, I from A

Legend



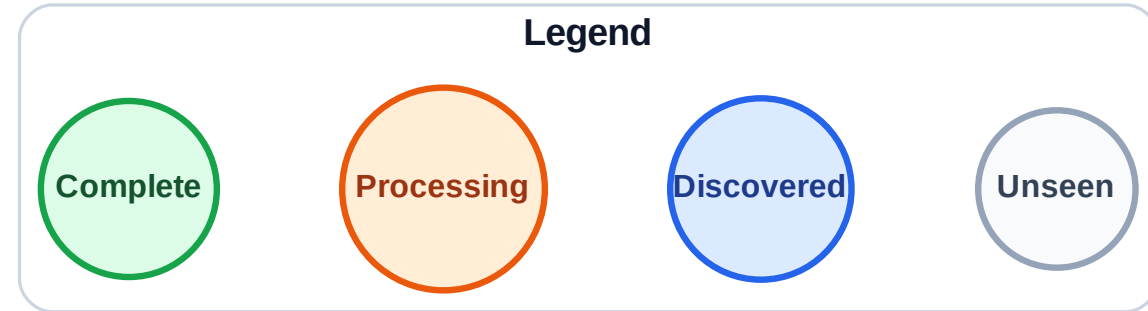
Queue: C → F → H → J → B → D → I

Visited: A, E



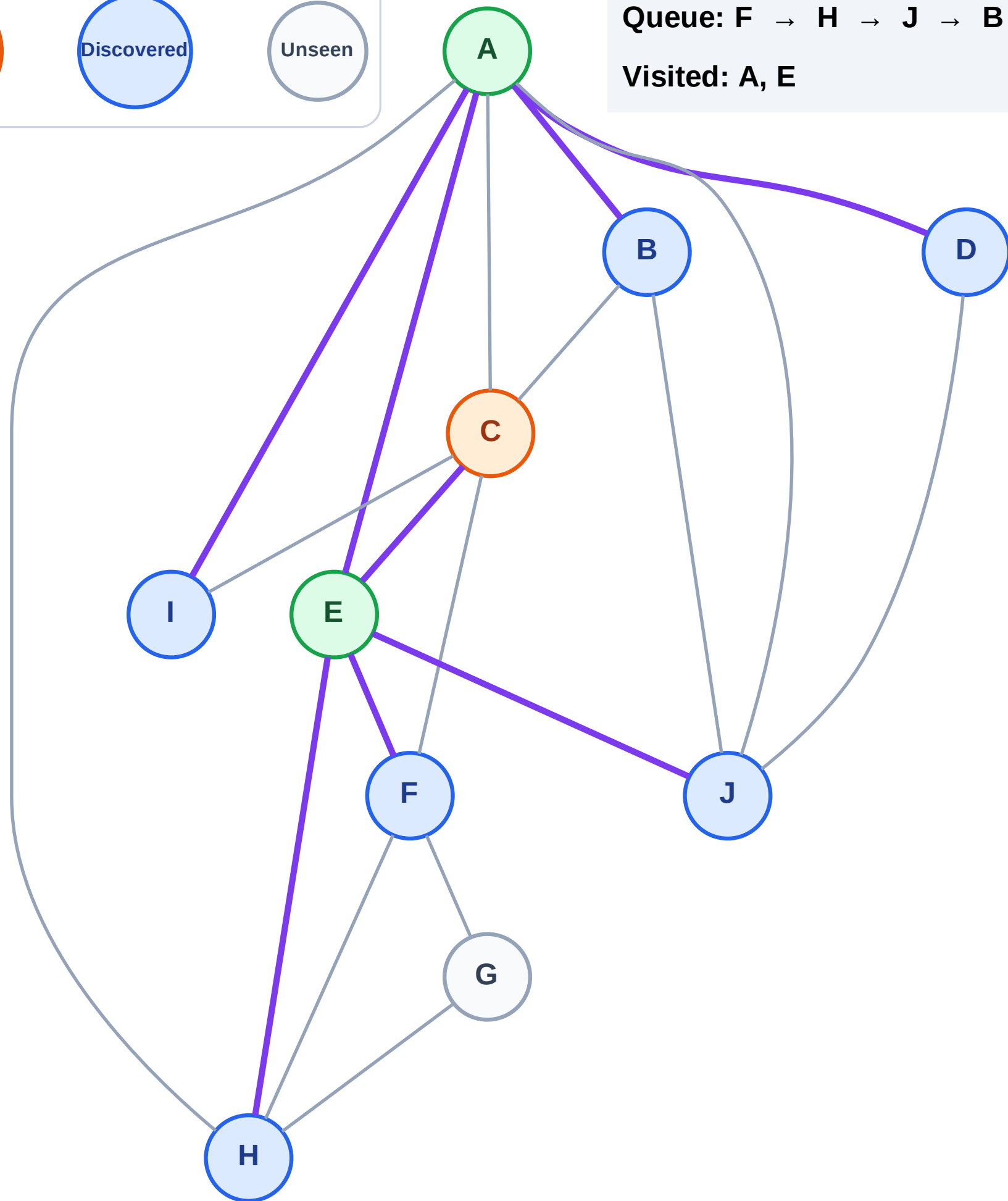
BFS Traversal

Step 6: Process node C



Queue: F → H → J → B → D → I

Visited: A, E



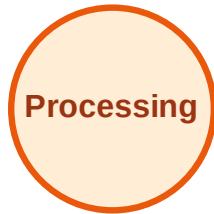
BFS Traversal

Step 7: No new neighbors from C

Legend



Complete



Processing

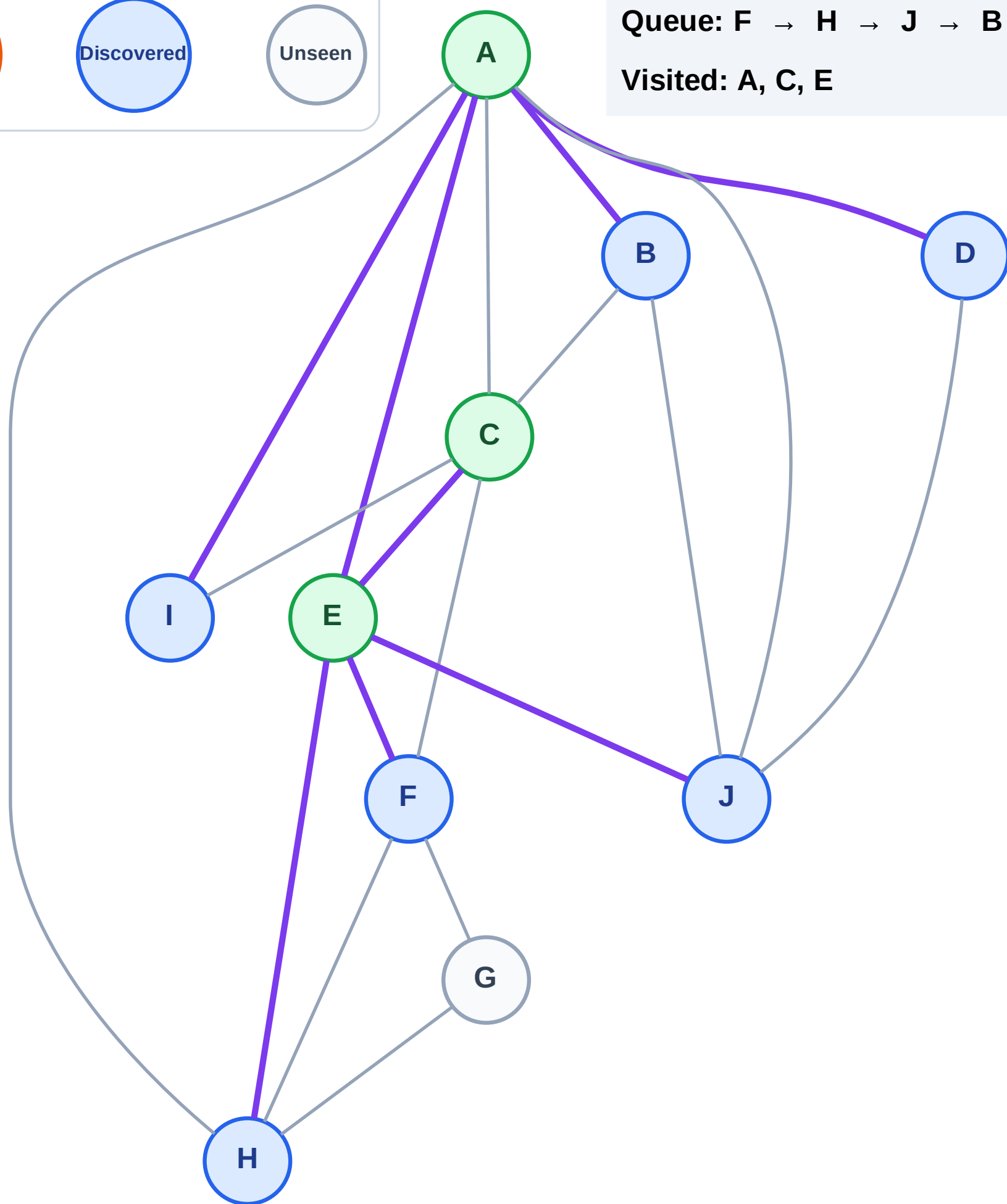


Discovered



Unseen

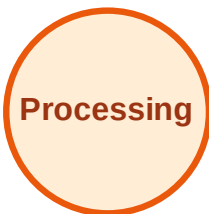
Queue: F → H → J → B → D → I
Visited: A, C, E



BFS Traversal

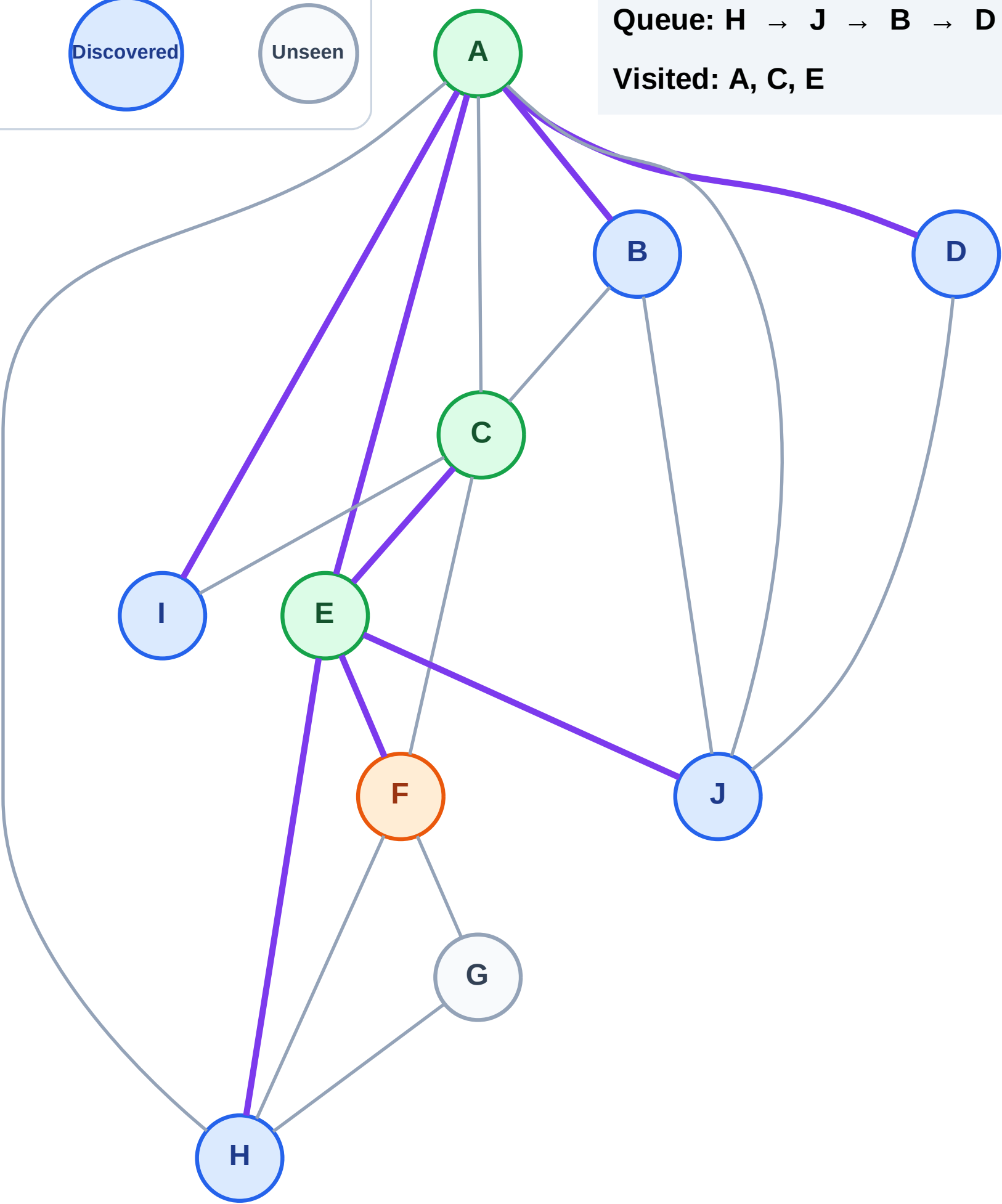
Step 8: Process node F

Legend



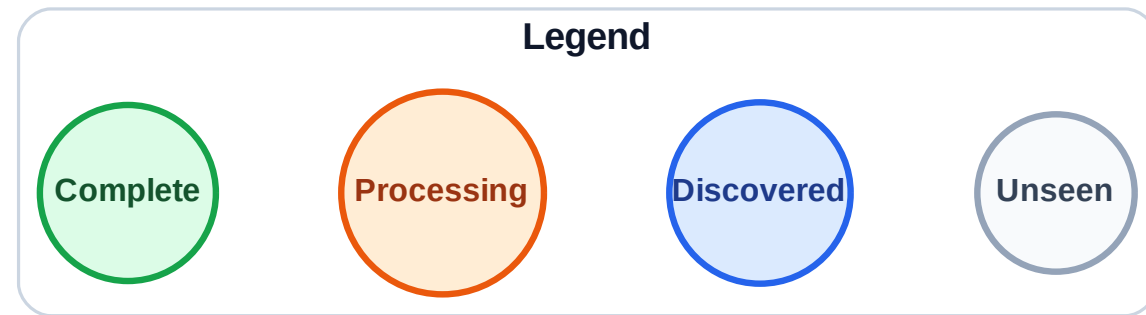
Queue: H → J → B → D → I

Visited: A, C, E



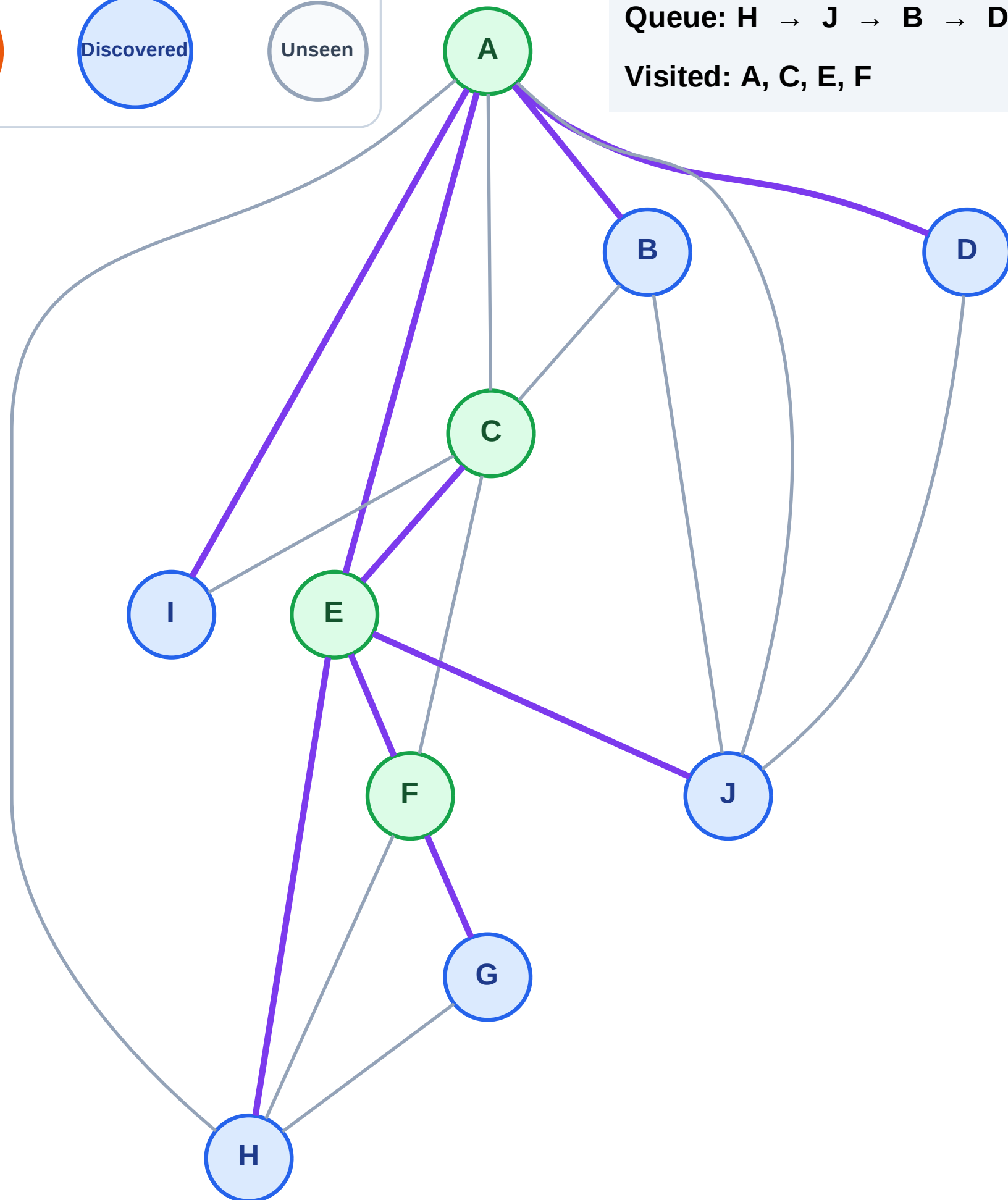
BFS Traversal

Step 9: Discover G from F



Queue: H → J → B → D → I → G

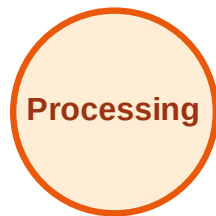
Visited: A, C, E, F



BFS Traversal

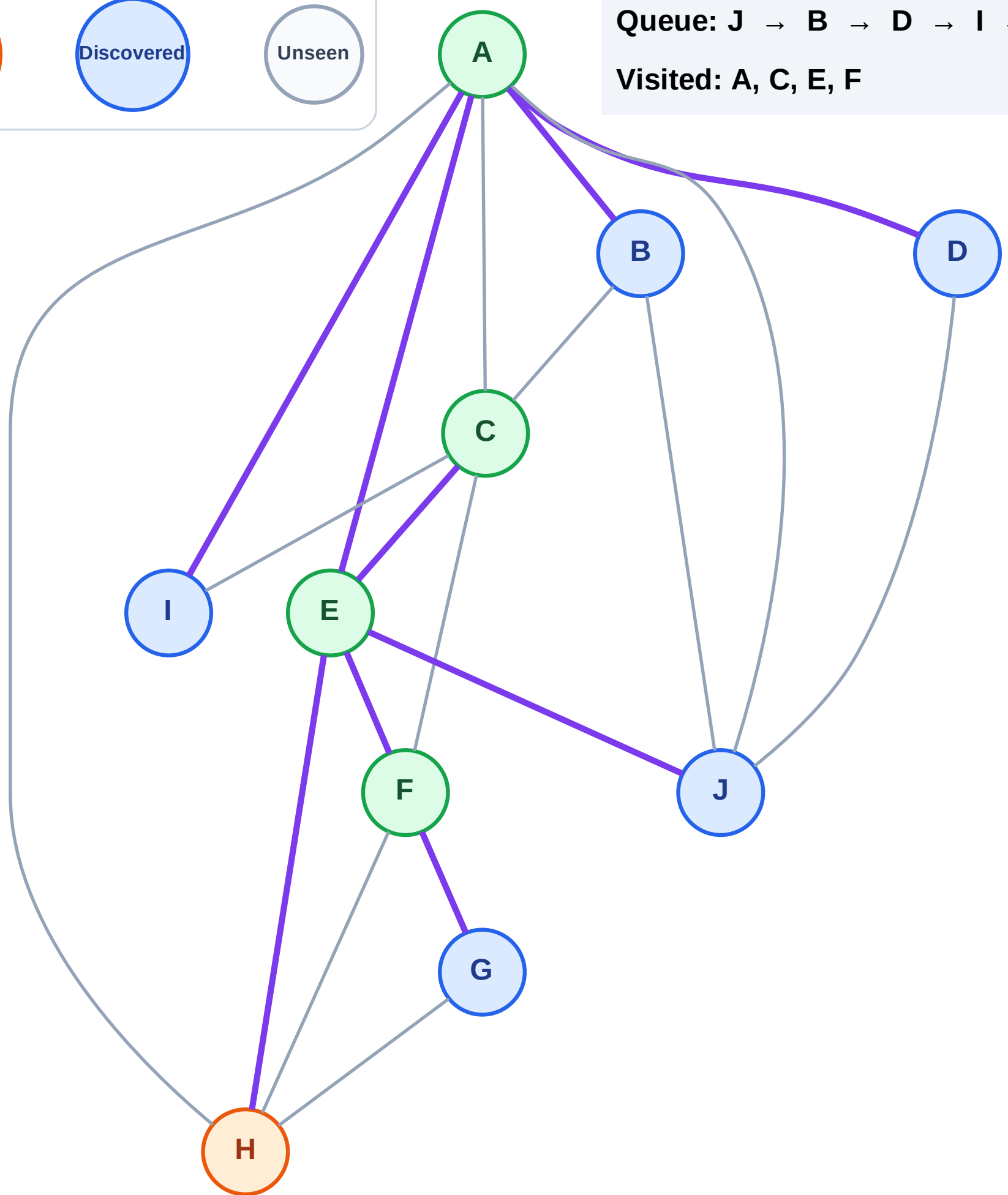
Step 10: Process node H

Legend



Queue: J → B → D → I → G

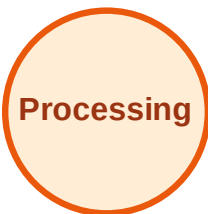
Visited: A, C, E, F



BFS Traversal

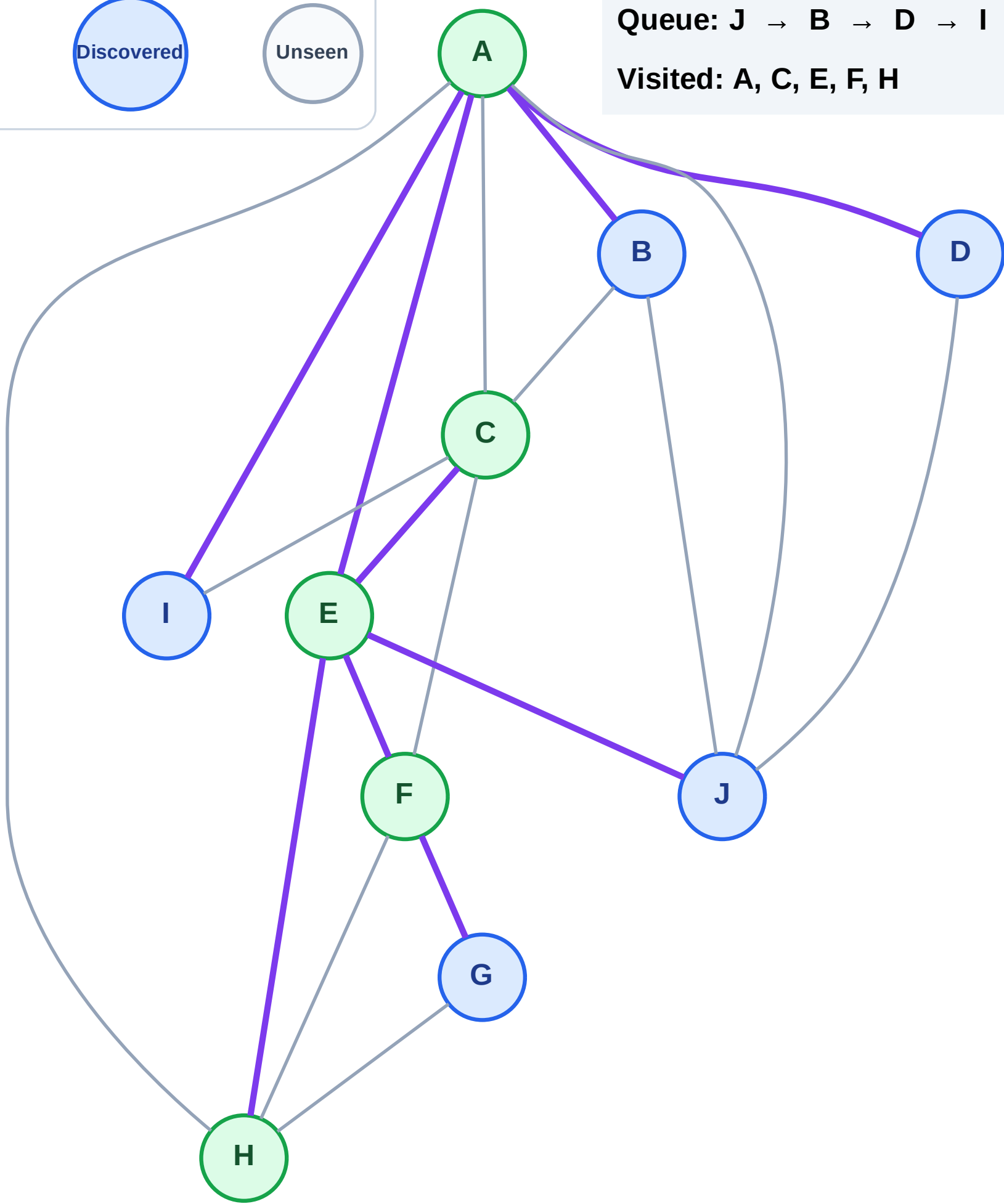
Step 11: No new neighbors from H

Legend



Queue: J → B → D → I → G

Visited: A, C, E, F, H



BFS Traversal

Step 12: Process node J

Legend

Complete

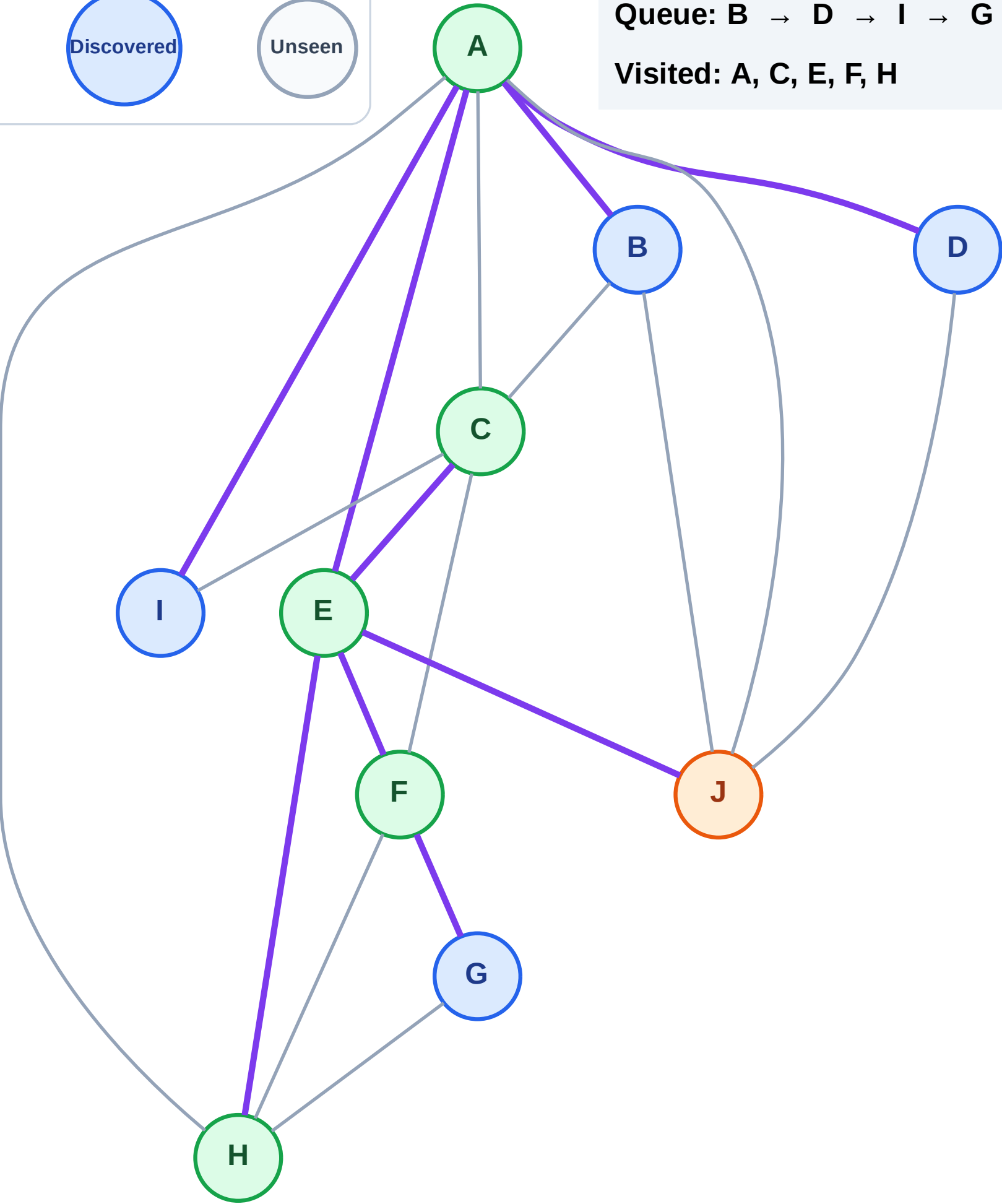
Processing

Discovered

Unseen

Queue: B → D → I → G

Visited: A, C, E, F, H



BFS Traversal

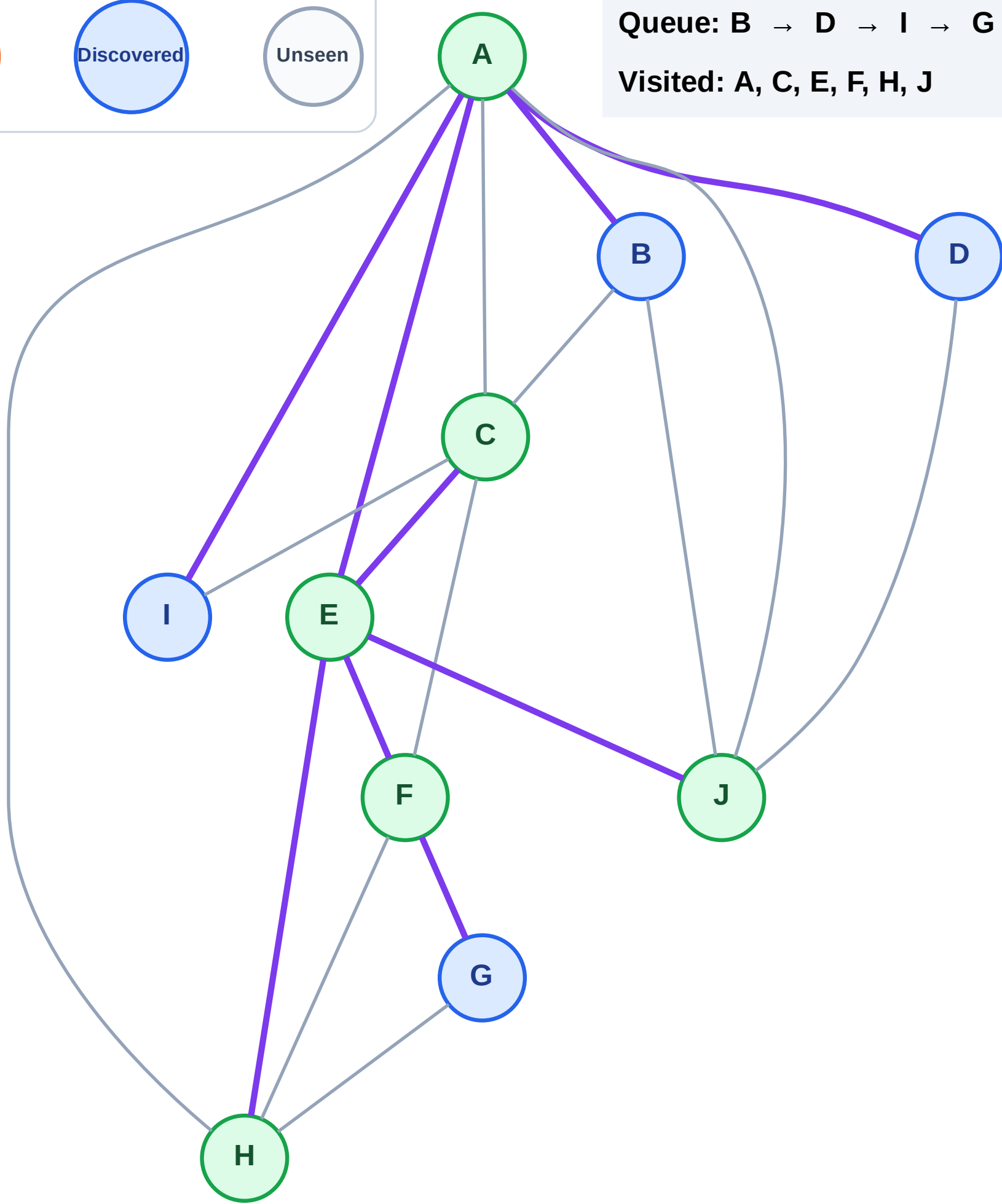
Step 13: No new neighbors from J

Legend



Queue: B → D → I → G

Visited: A, C, E, F, H, J



BFS Traversal

Step 14: Process node B

Legend

Complete

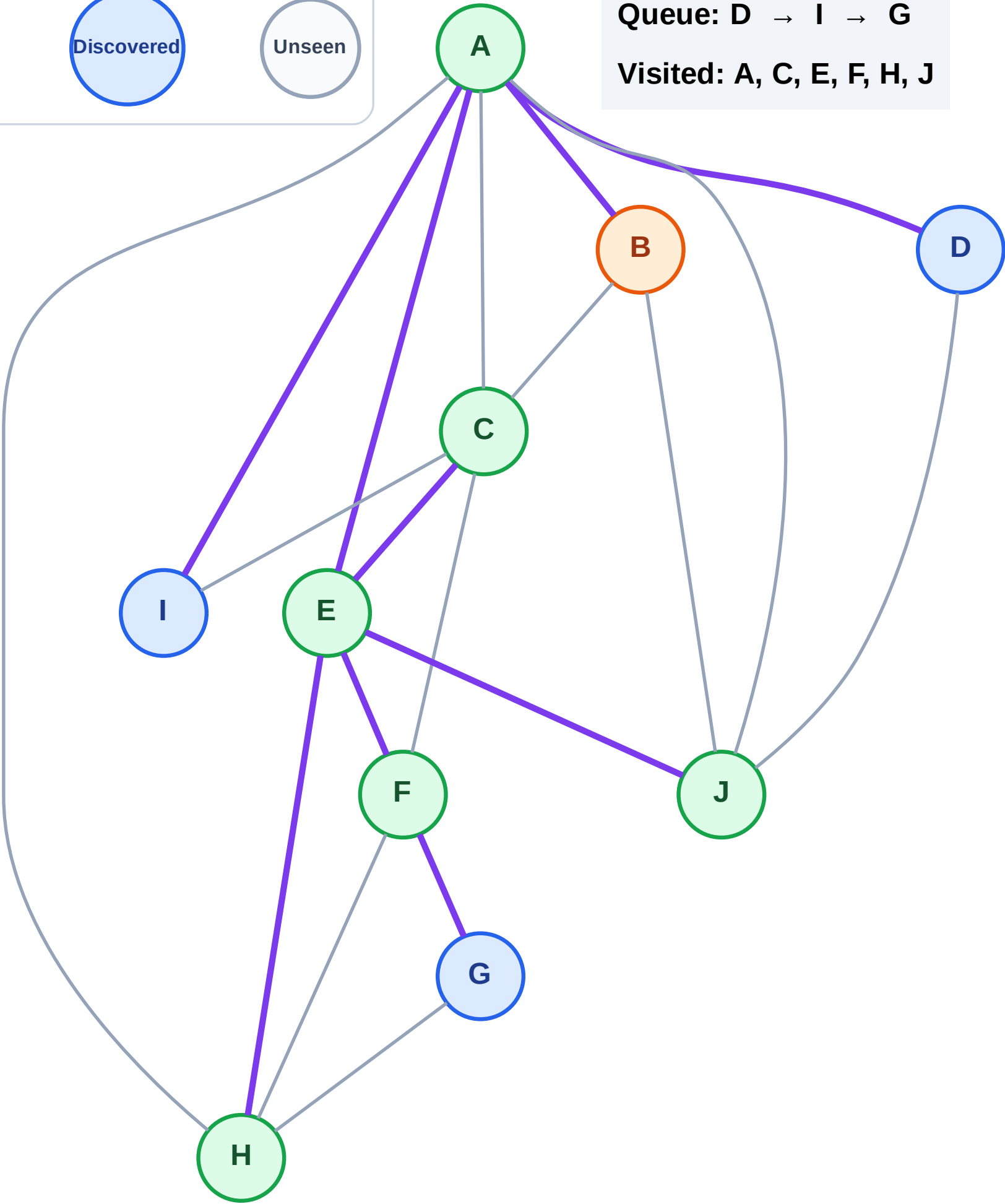
Processing

Discovered

Unseen

Queue: D → I → G

Visited: A, C, E, F, H, J



BFS Traversal

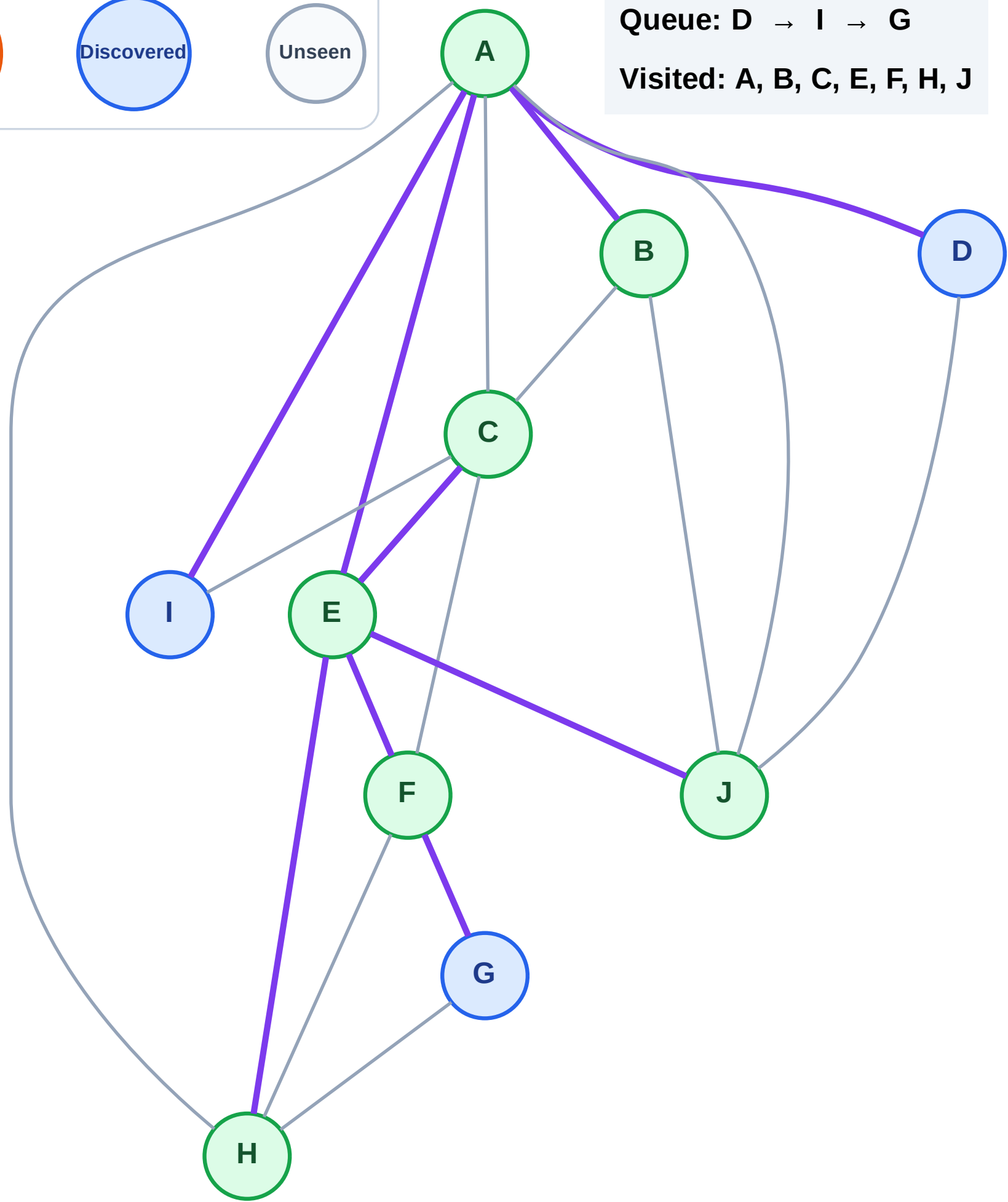
Step 15: No new neighbors from B

Legend



Queue: D → I → G

Visited: A, B, C, E, F, H, J



BFS Traversal

Step 16: Process node D

Legend

Complete

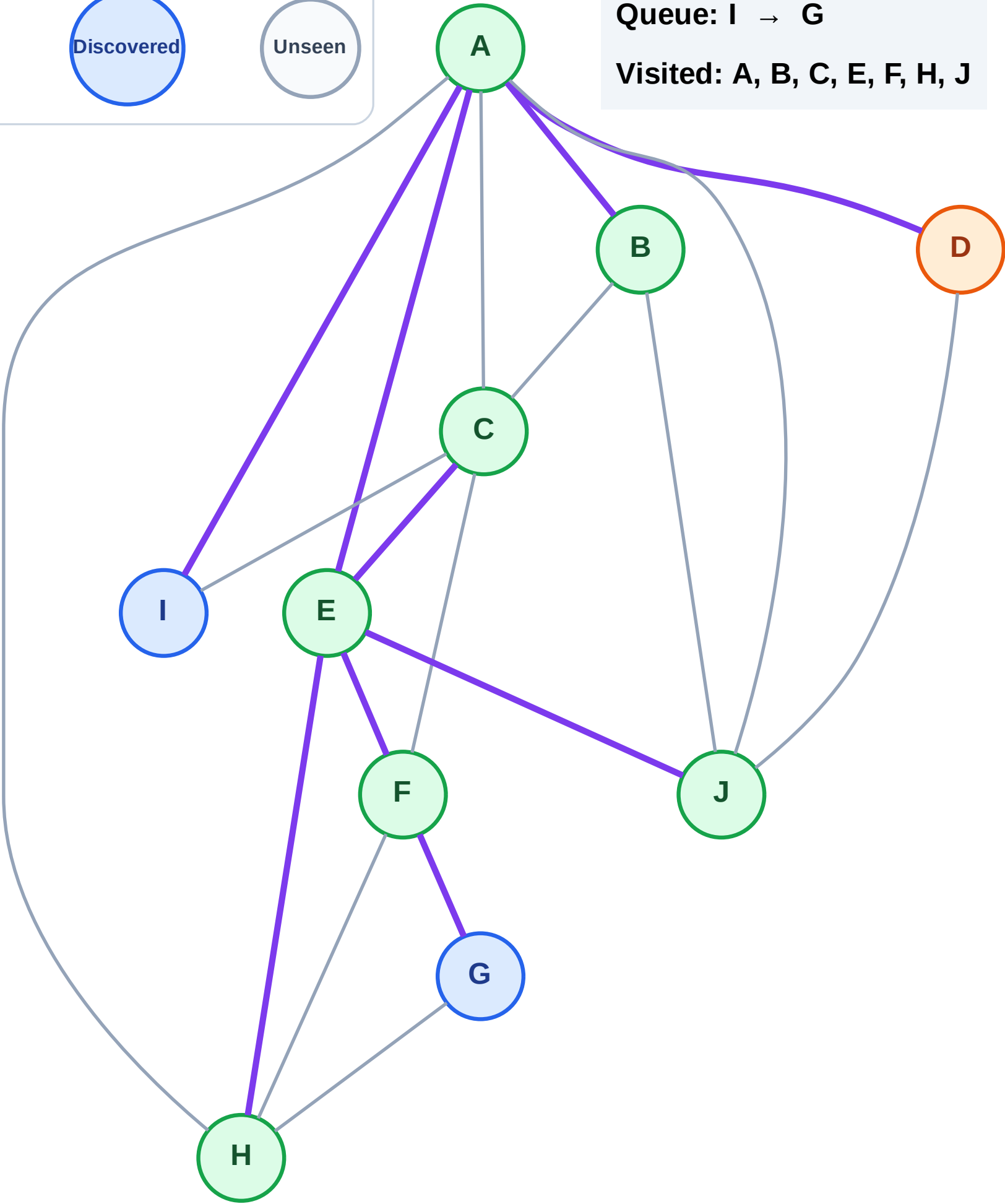
Processing

Discovered

Unseen

Queue: I → G

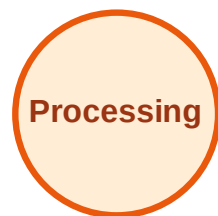
Visited: A, B, C, E, F, H, J



BFS Traversal

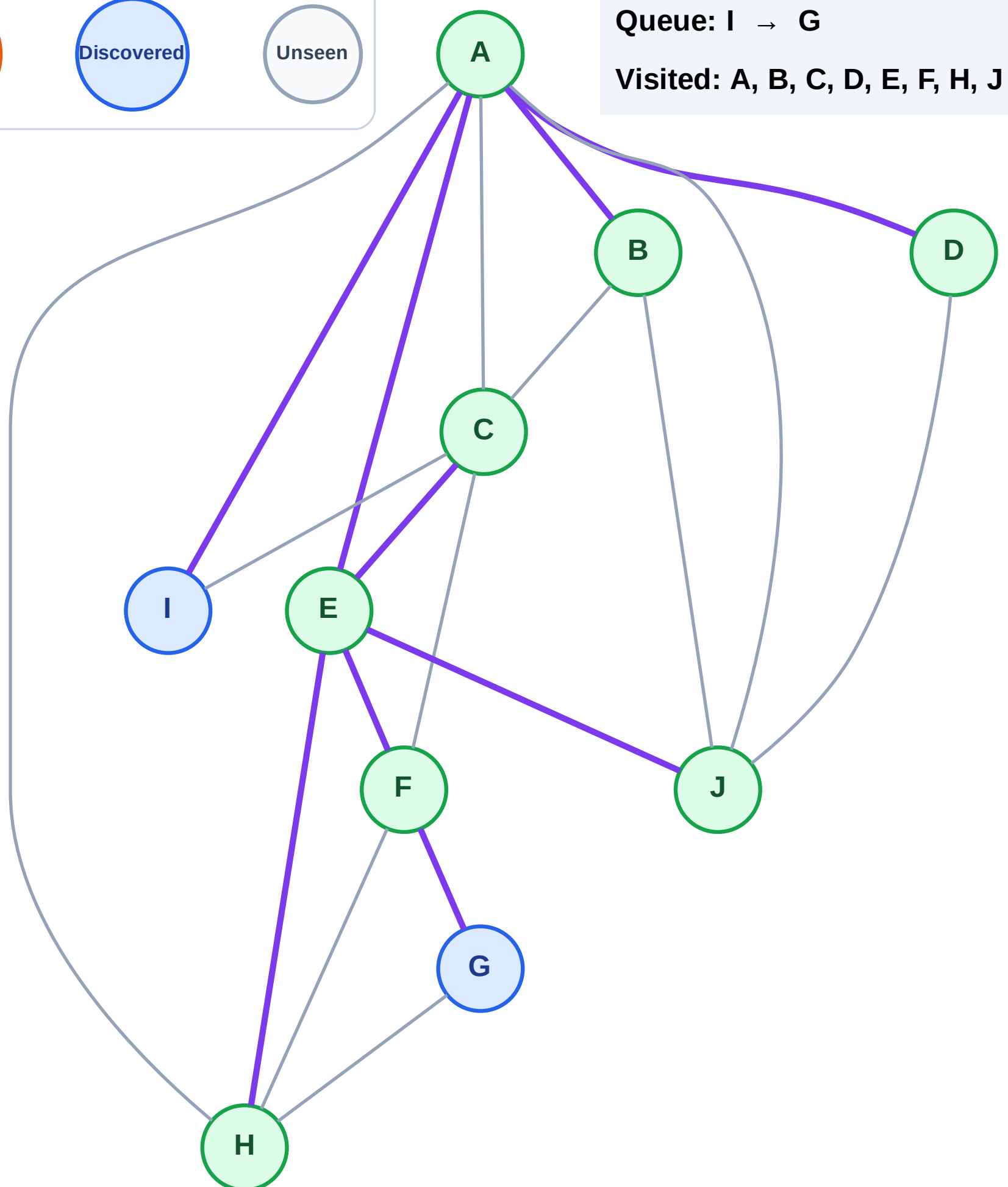
Step 17: No new neighbors from D

Legend



Queue: I → G

Visited: A, B, C, D, E, F, H, J



BFS Traversal

Step 18: Process node I

Legend

Complete

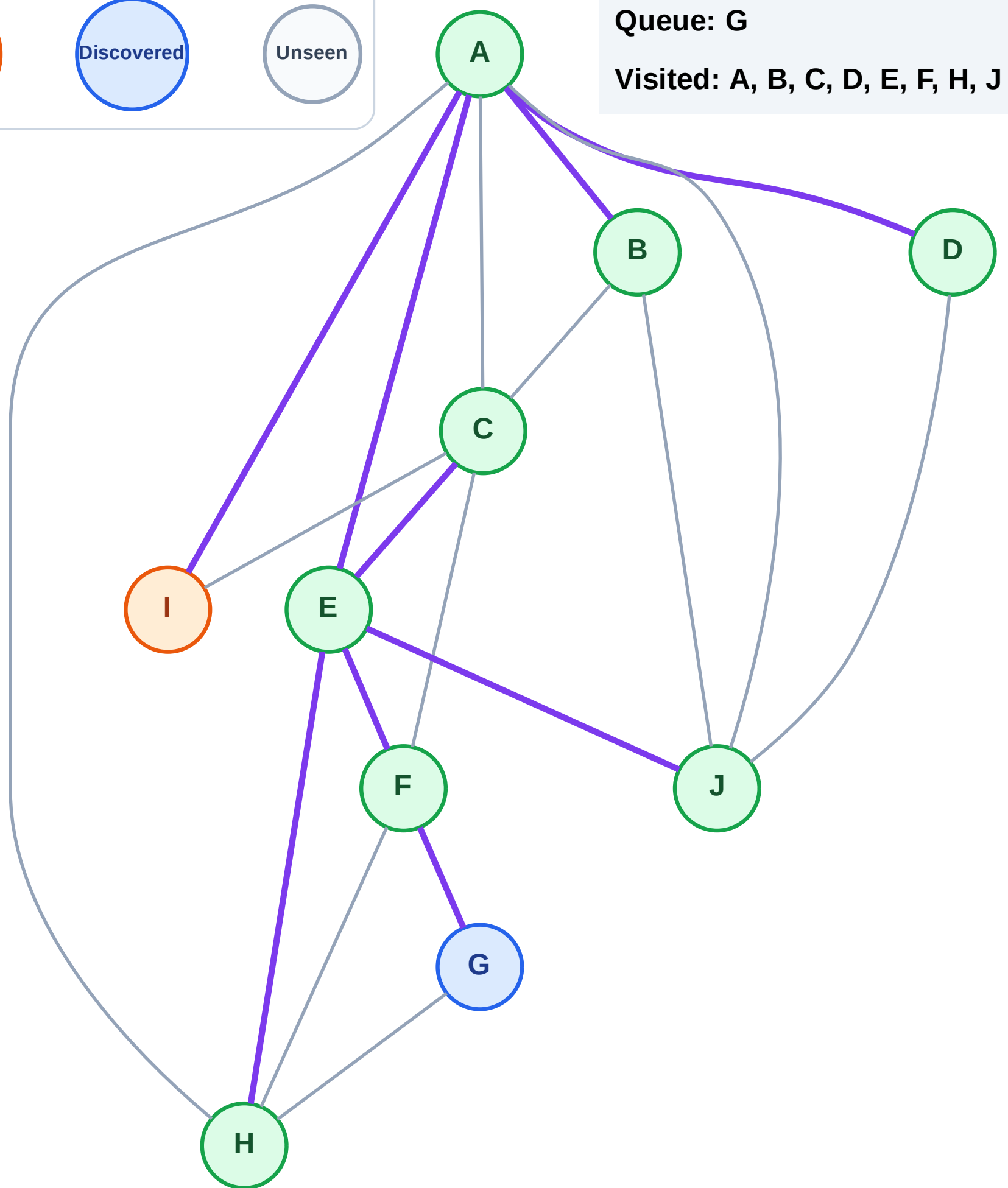
Processing

Discovered

Unseen

Queue: G

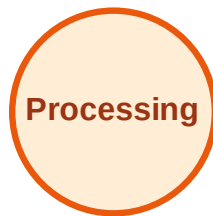
Visited: A, B, C, D, E, F, H, J



BFS Traversal

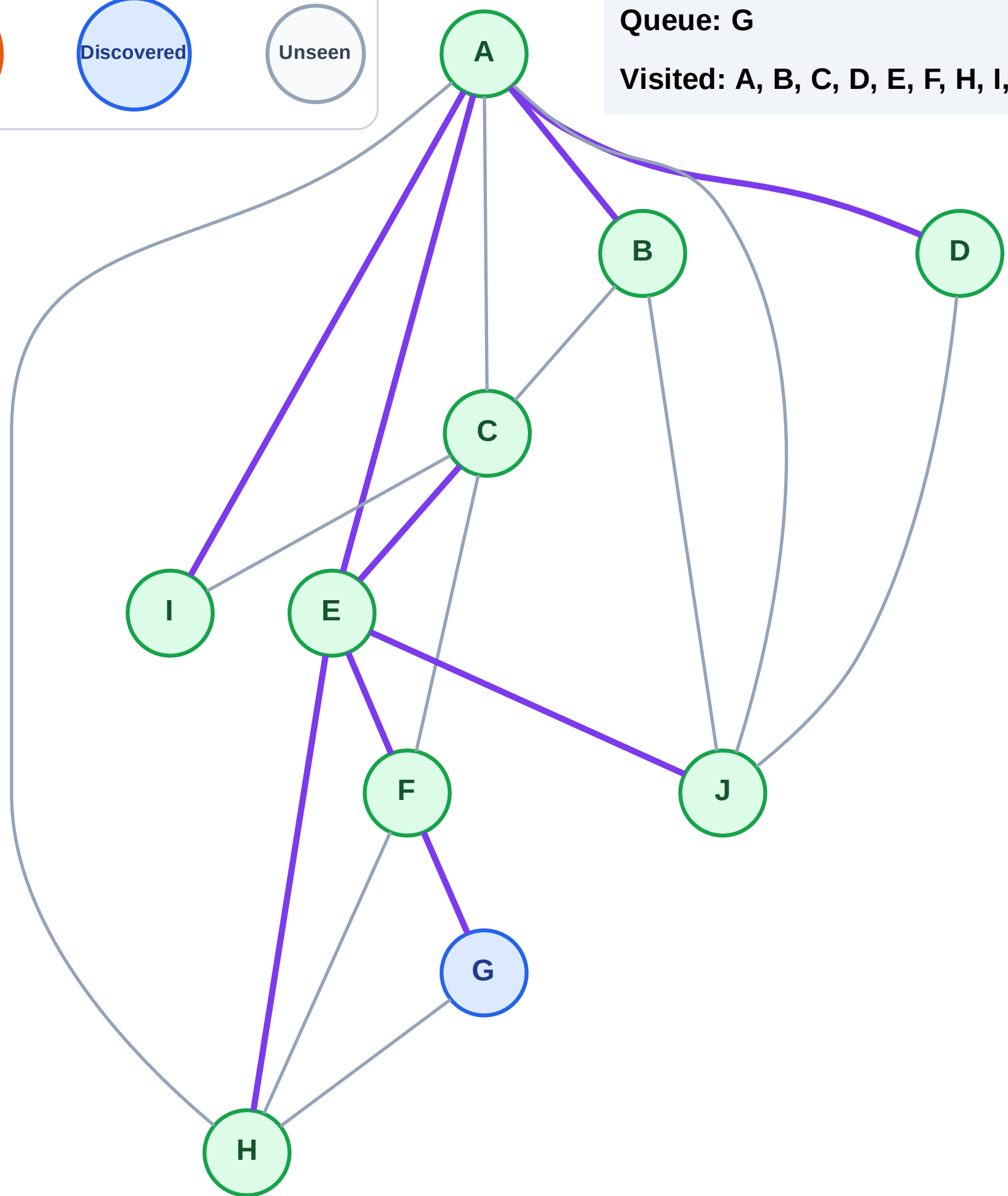
Step 19: No new neighbors from I

Legend



Queue: G

Visited: A, B, C, D, E, F, H, I, J



BFS Traversal

Step 20: Process node G

Legend

Complete

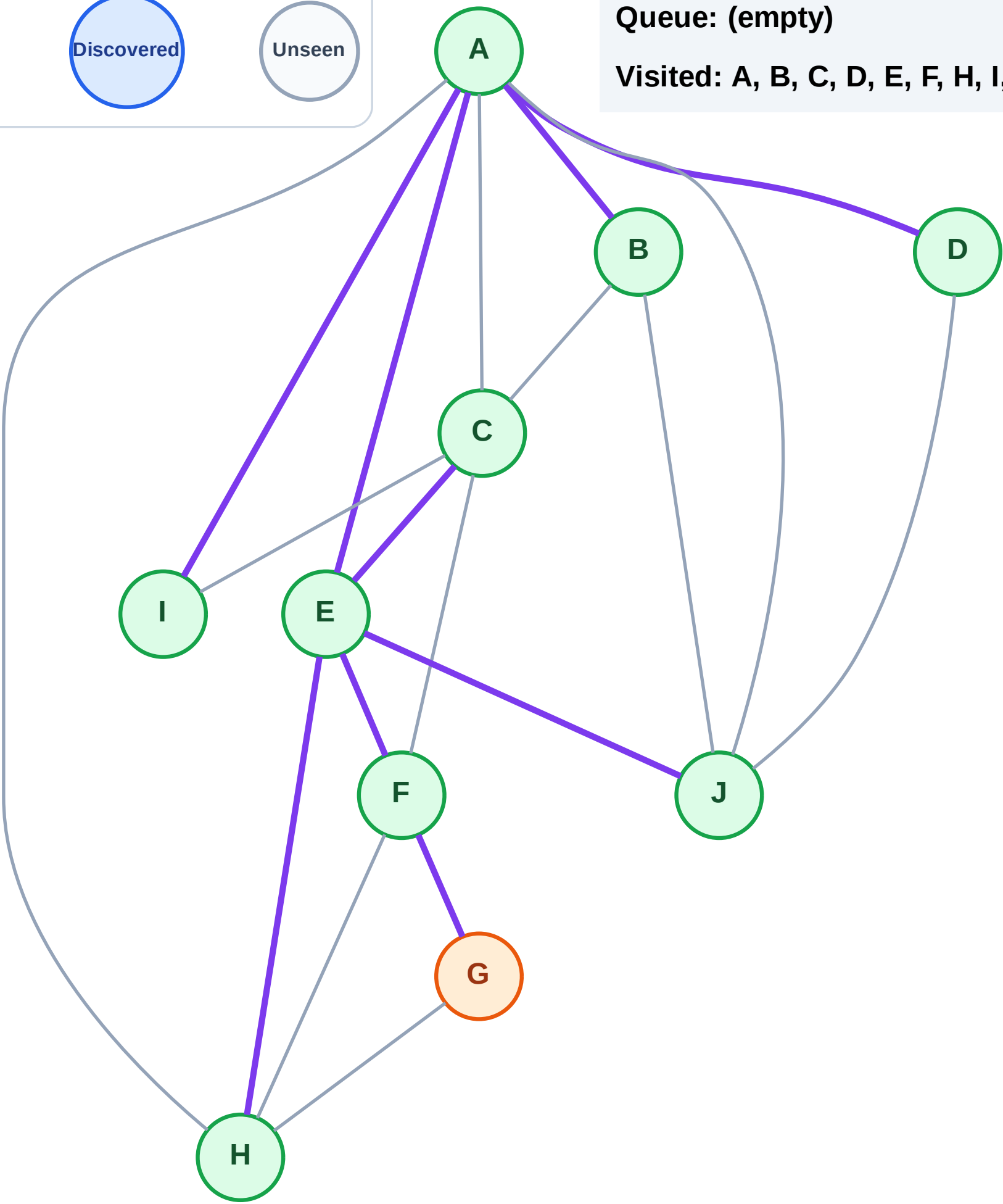
Processing

Discovered

Unseen

Queue: (empty)

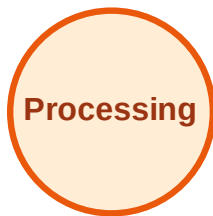
Visited: A, B, C, D, E, F, H, I, J



BFS Traversal

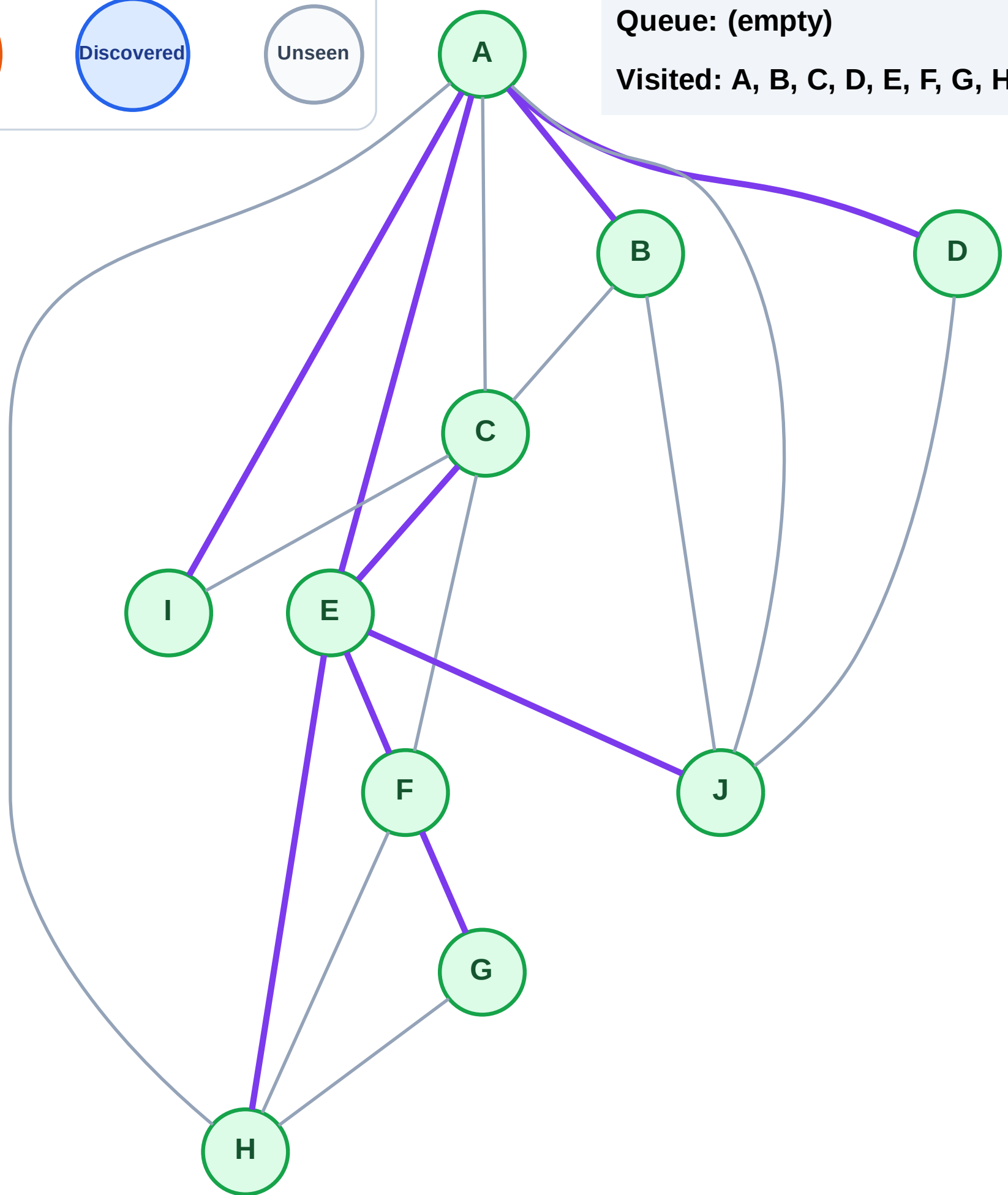
Step 21: No new neighbors from G

Legend



Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

Step 1: Start at E

Legend

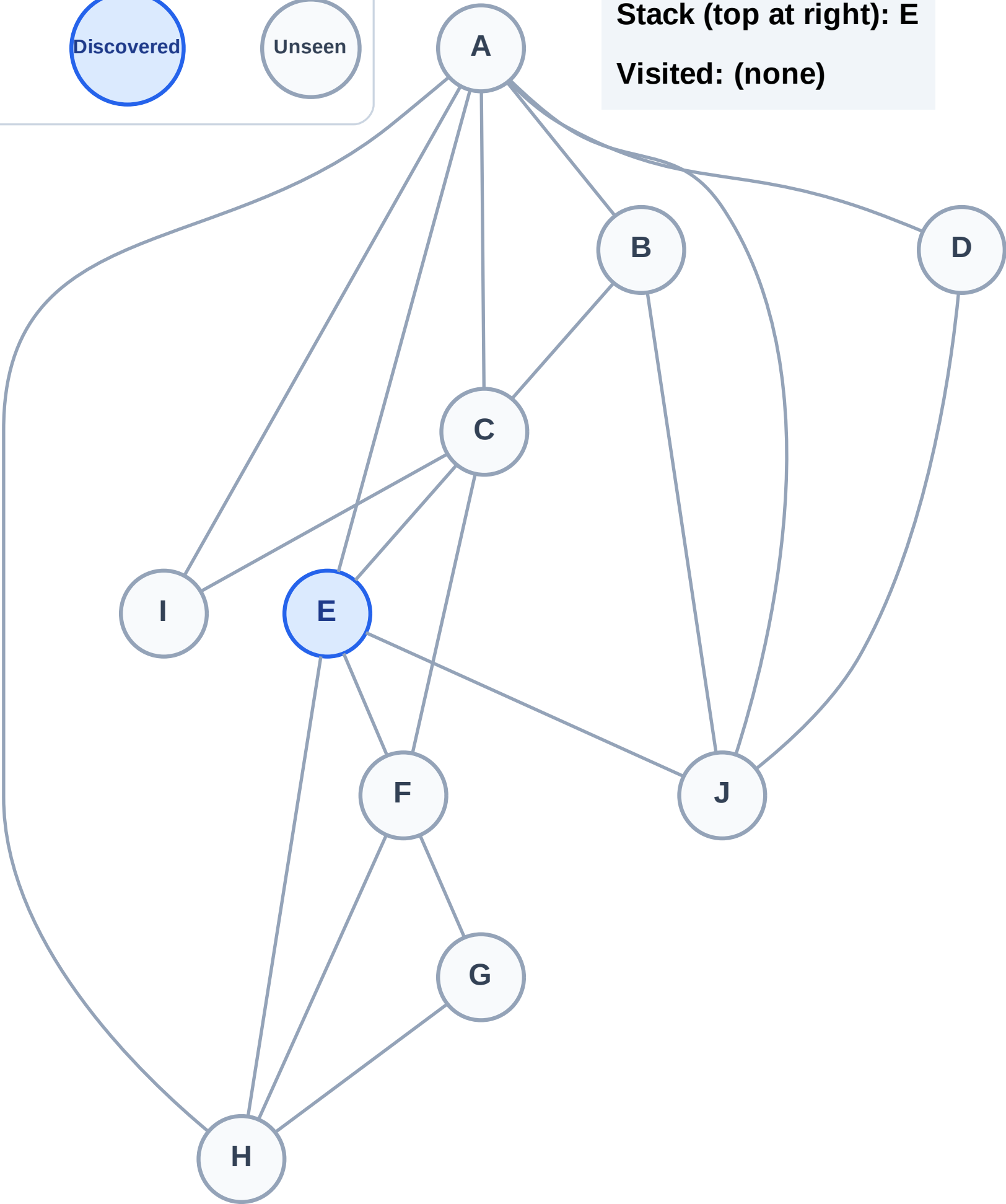
Complete

Processing

Discovered

Unseen

Stack (top at right): E
Visited: (none)



DFS Traversal

Step 2: Process node E

Legend

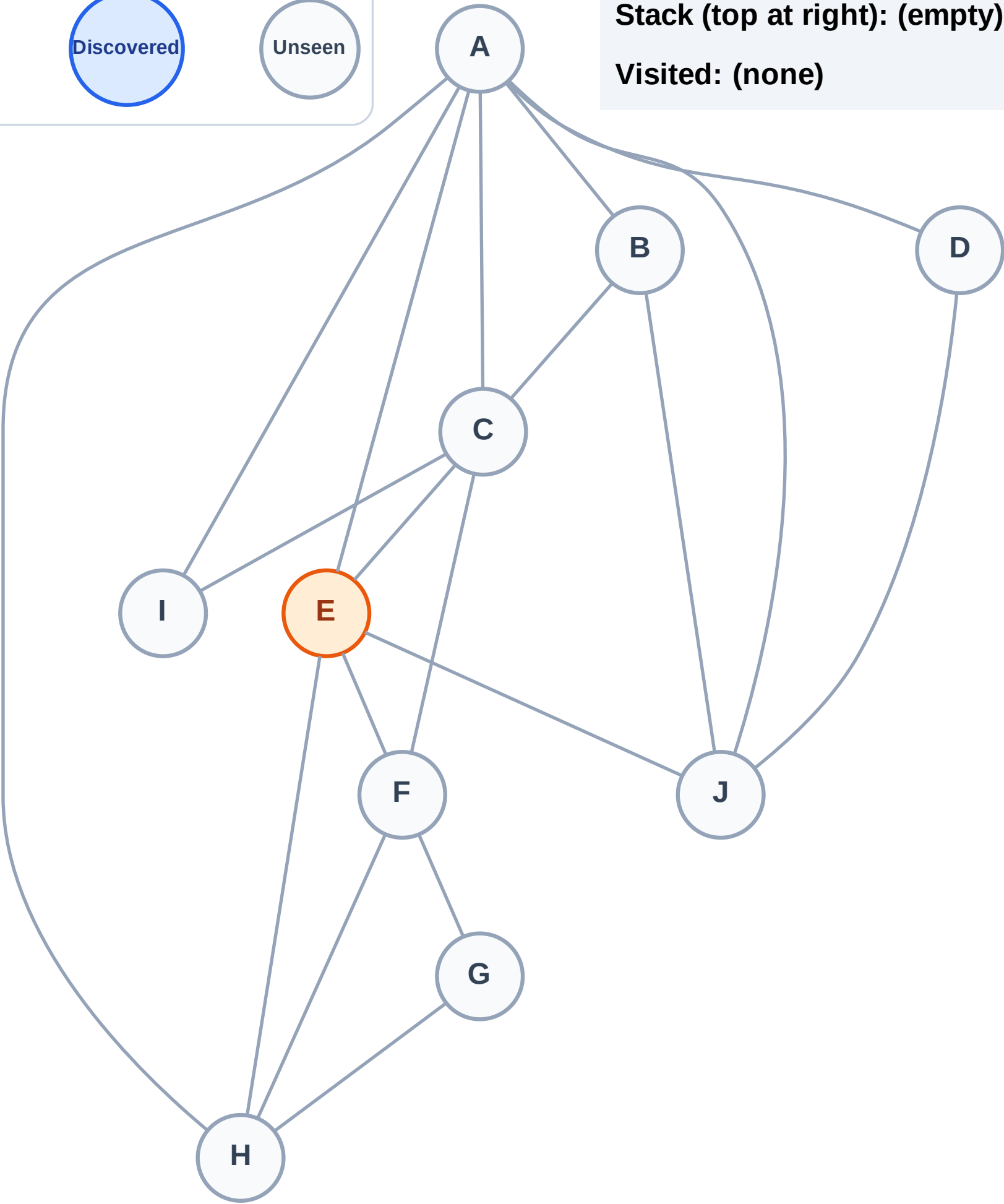
Complete

Processing

Discovered

Unseen

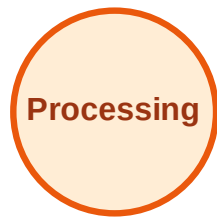
Stack (top at right): (empty)
Visited: (none)



DFS Traversal

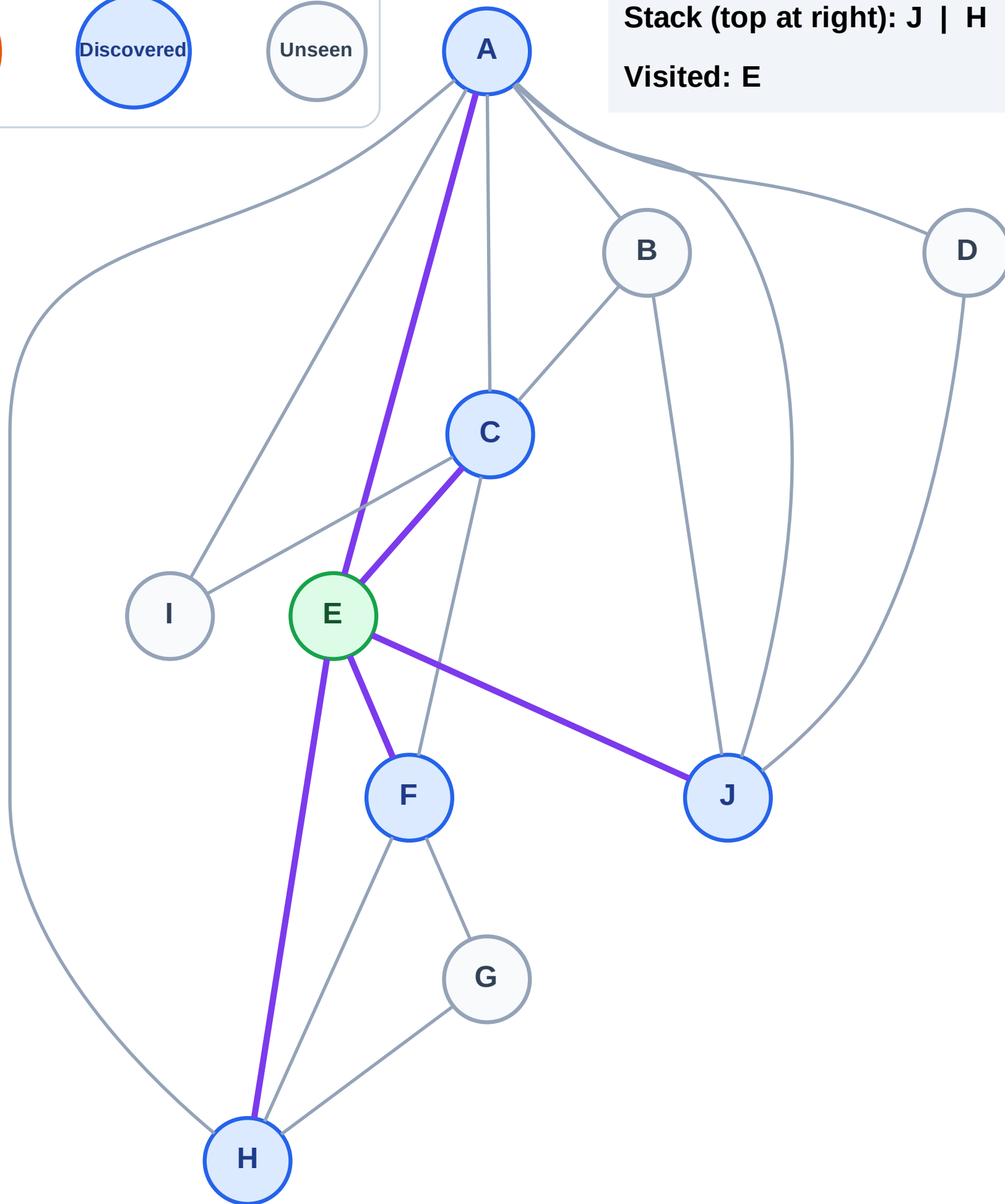
Step 3: Discover J, H, F, C, A from E

Legend



Stack (top at right): J | H | F | C | A

Visited: E



DFS Traversal

Step 4: Process node A

Legend

Complete

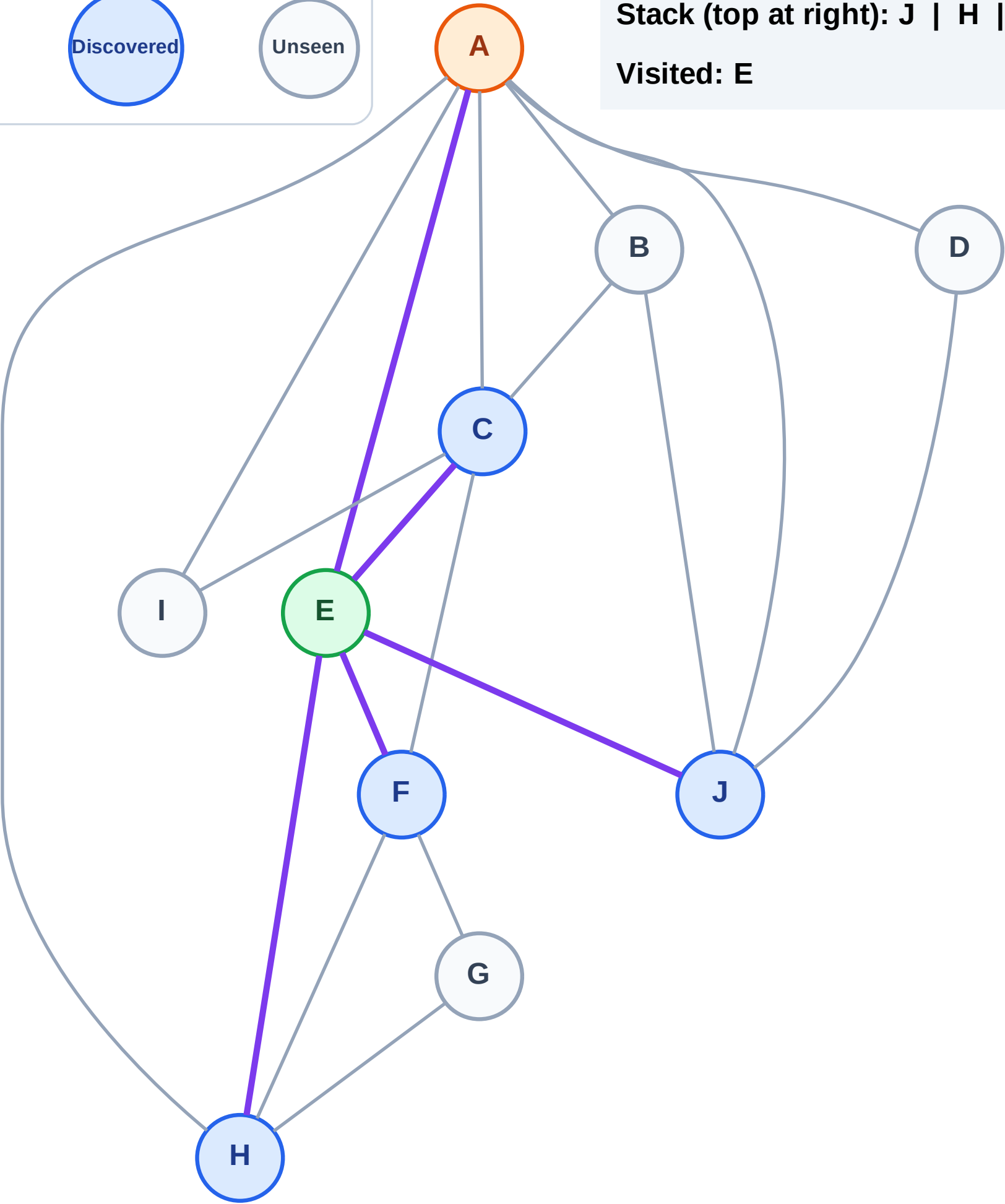
Processing

Discovered

Unseen

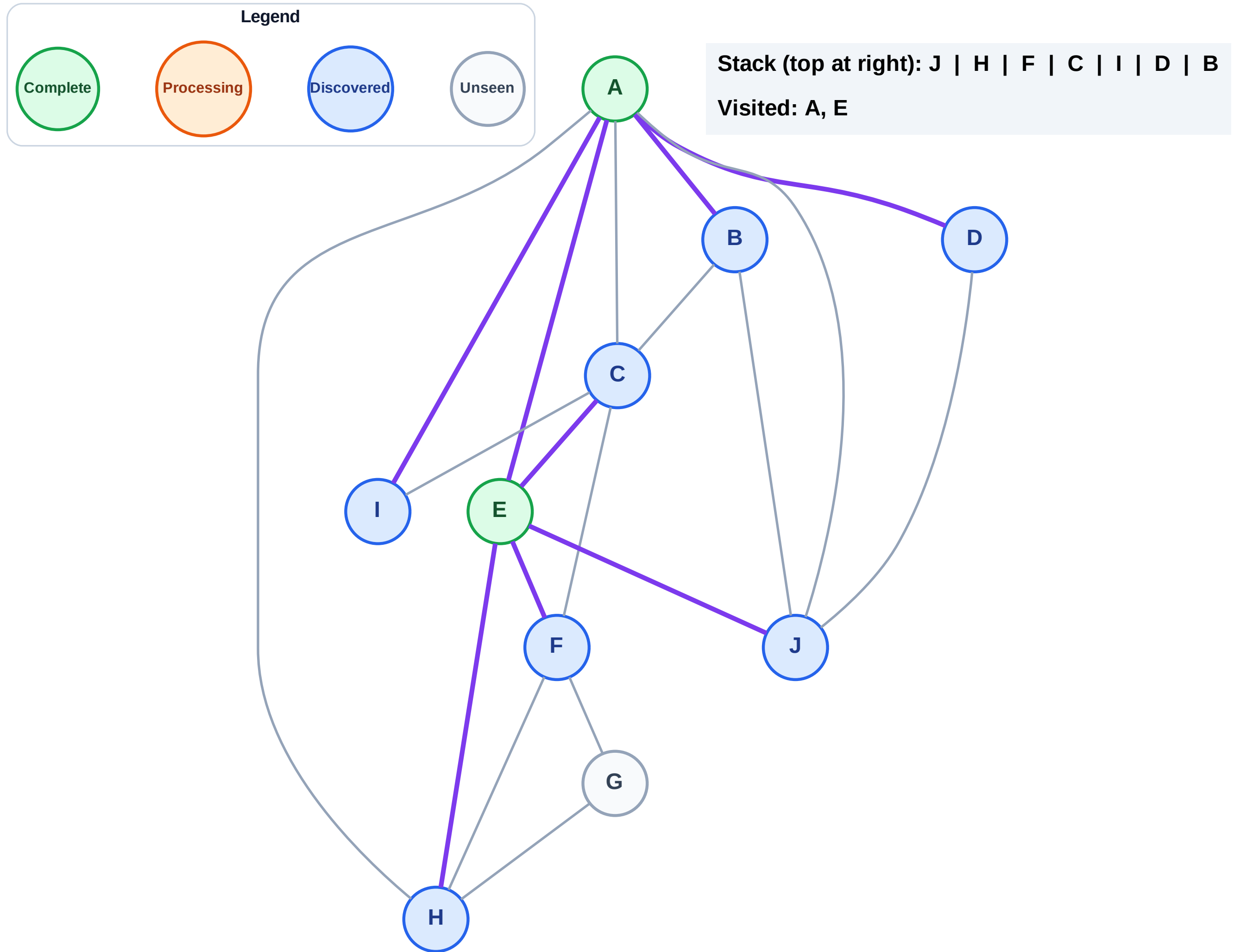
Stack (top at right): J | H | F | C

Visited: E



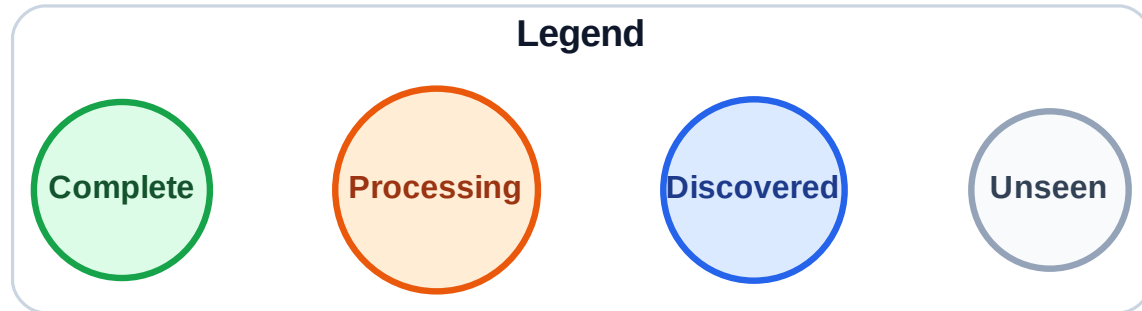
Step 5: Discover I, D, B from A

Step 5: Discover I, D, B from A



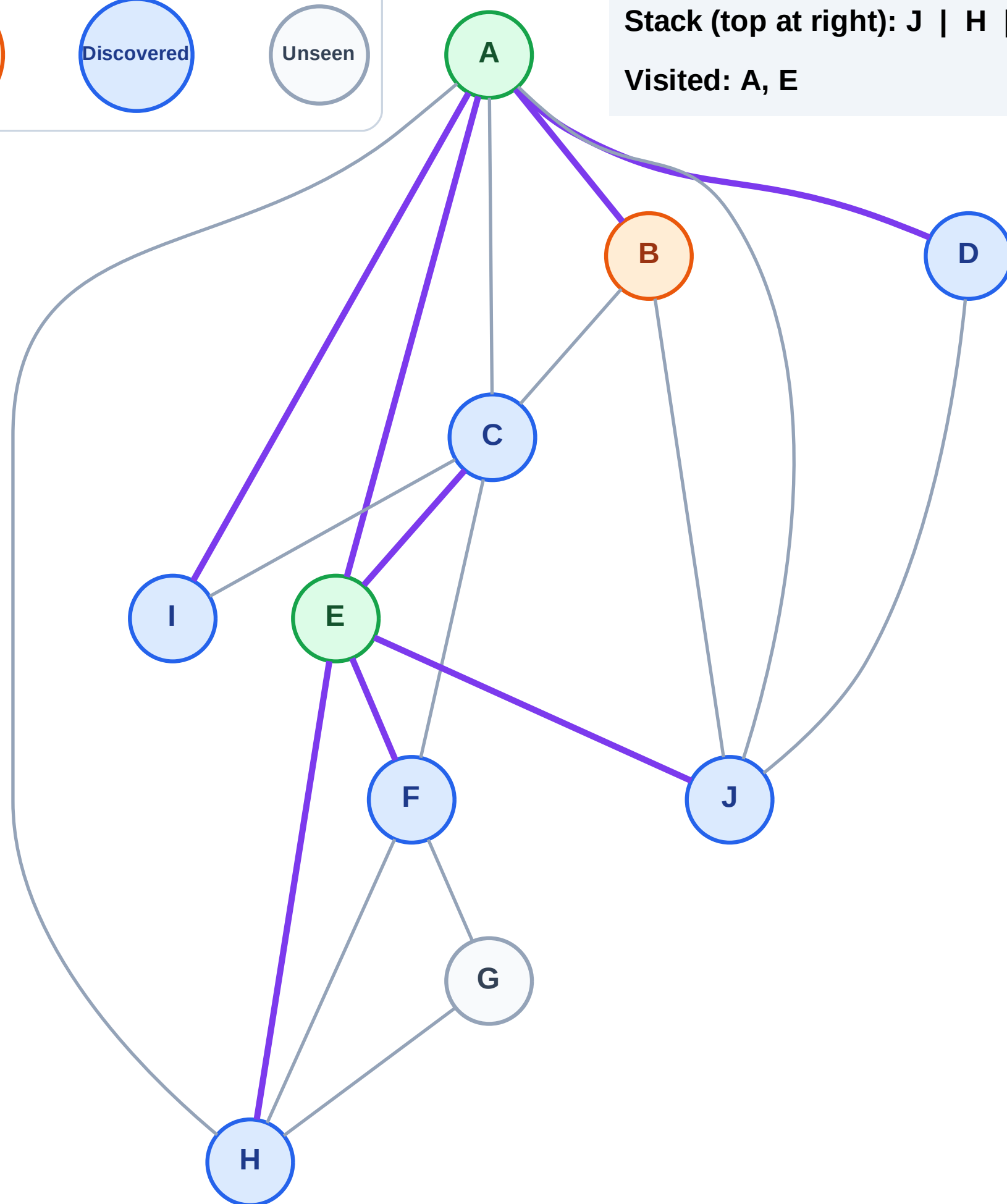
DFS Traversal

Step 6: Process node B



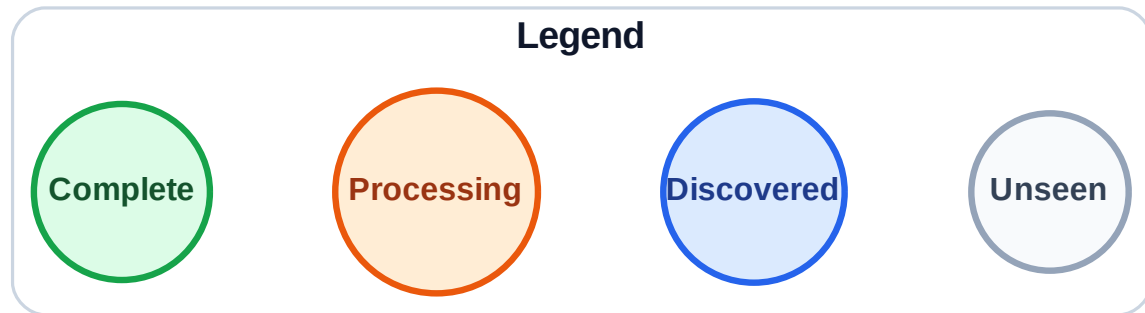
Stack (top at right): J | H | F | C | I | D

Visited: A, E

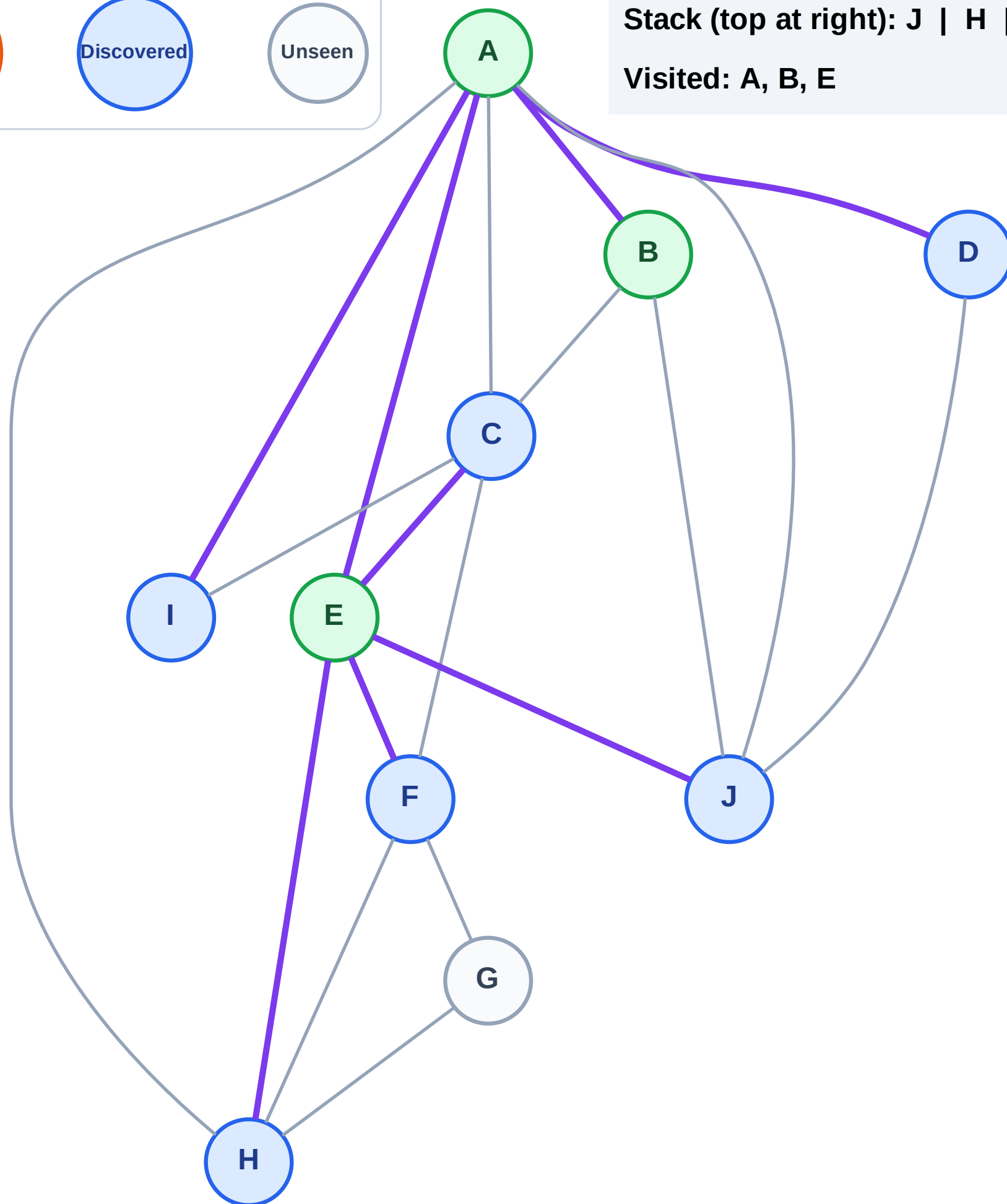


DFS Traversal

Step 7: No new neighbors from B



Stack (top at right): J | H | F | C | I | D
Visited: A, B, E

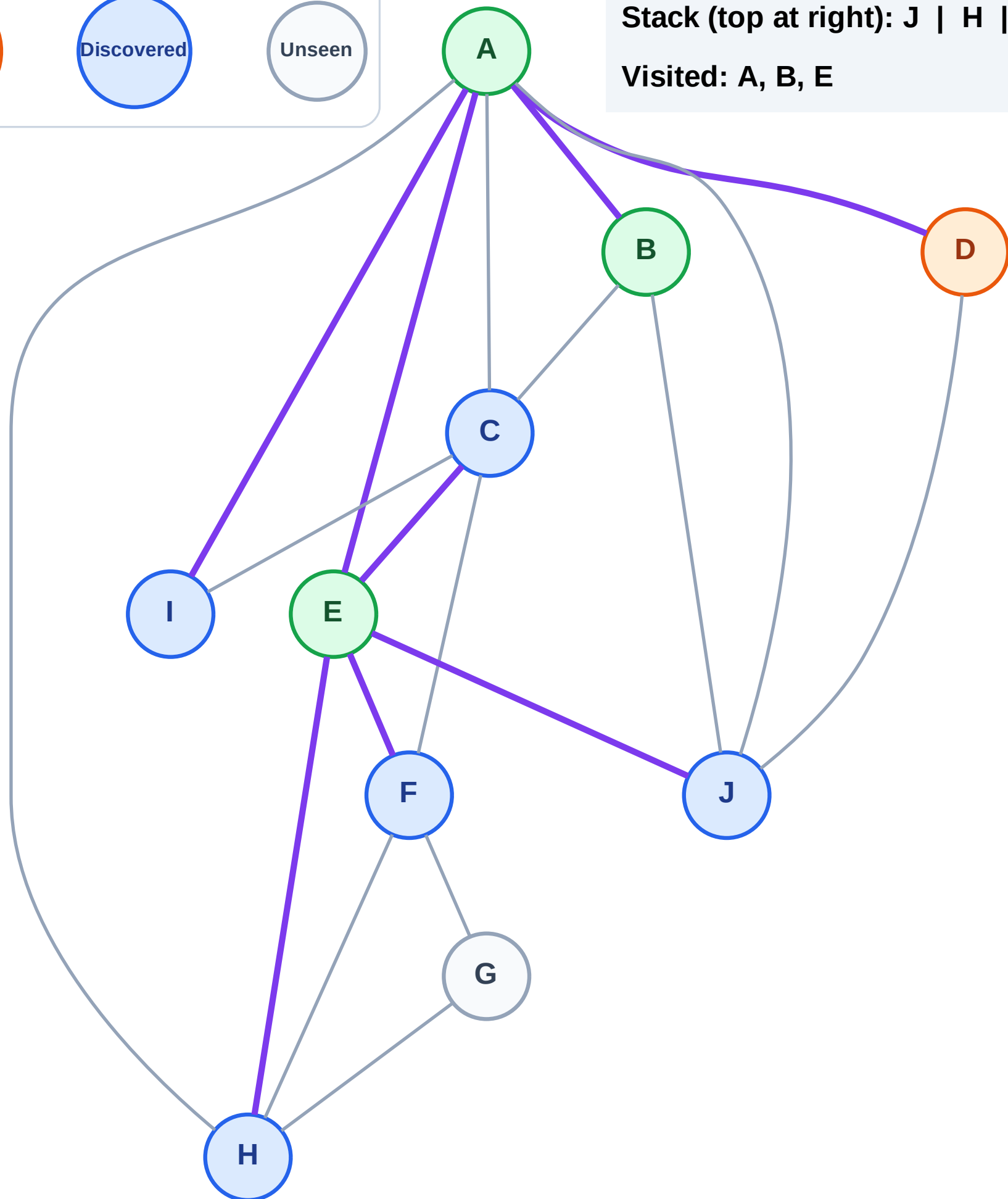
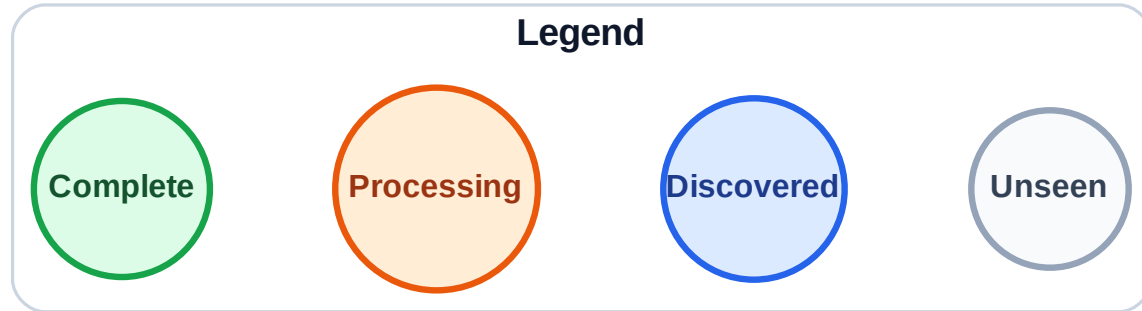


DFS Traversal

Step 8: Process node D

Stack (top at right): J | H | F | C | I

Visited: A, B, E



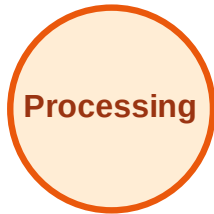
DFS Traversal

Step 9: No new neighbors from D

Legend



Complete



Processing



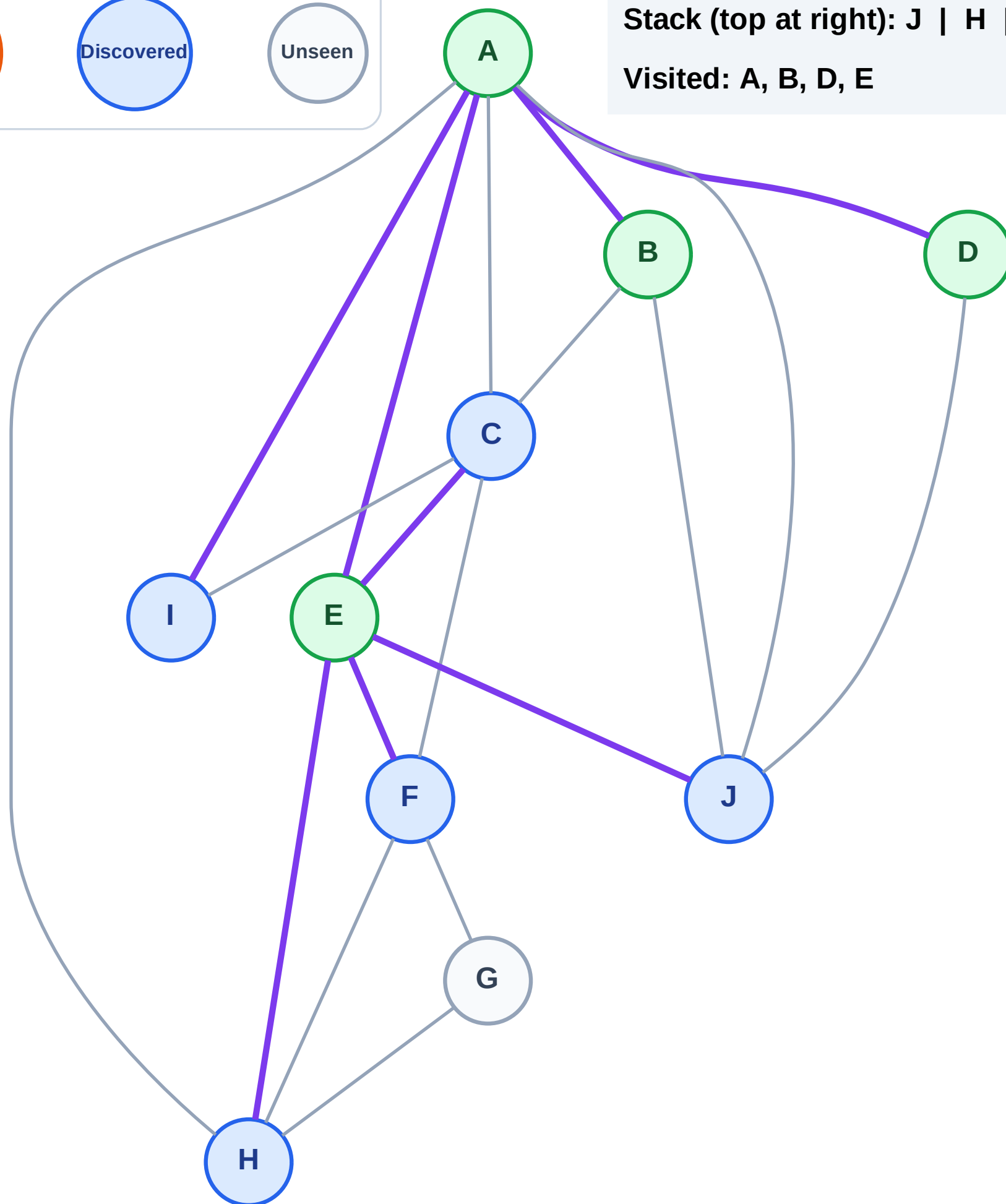
Discovered



Unseen

Stack (top at right): J | H | F | C | I

Visited: A, B, D, E



DFS Traversal

Step 10: Process node I

Legend

Complete

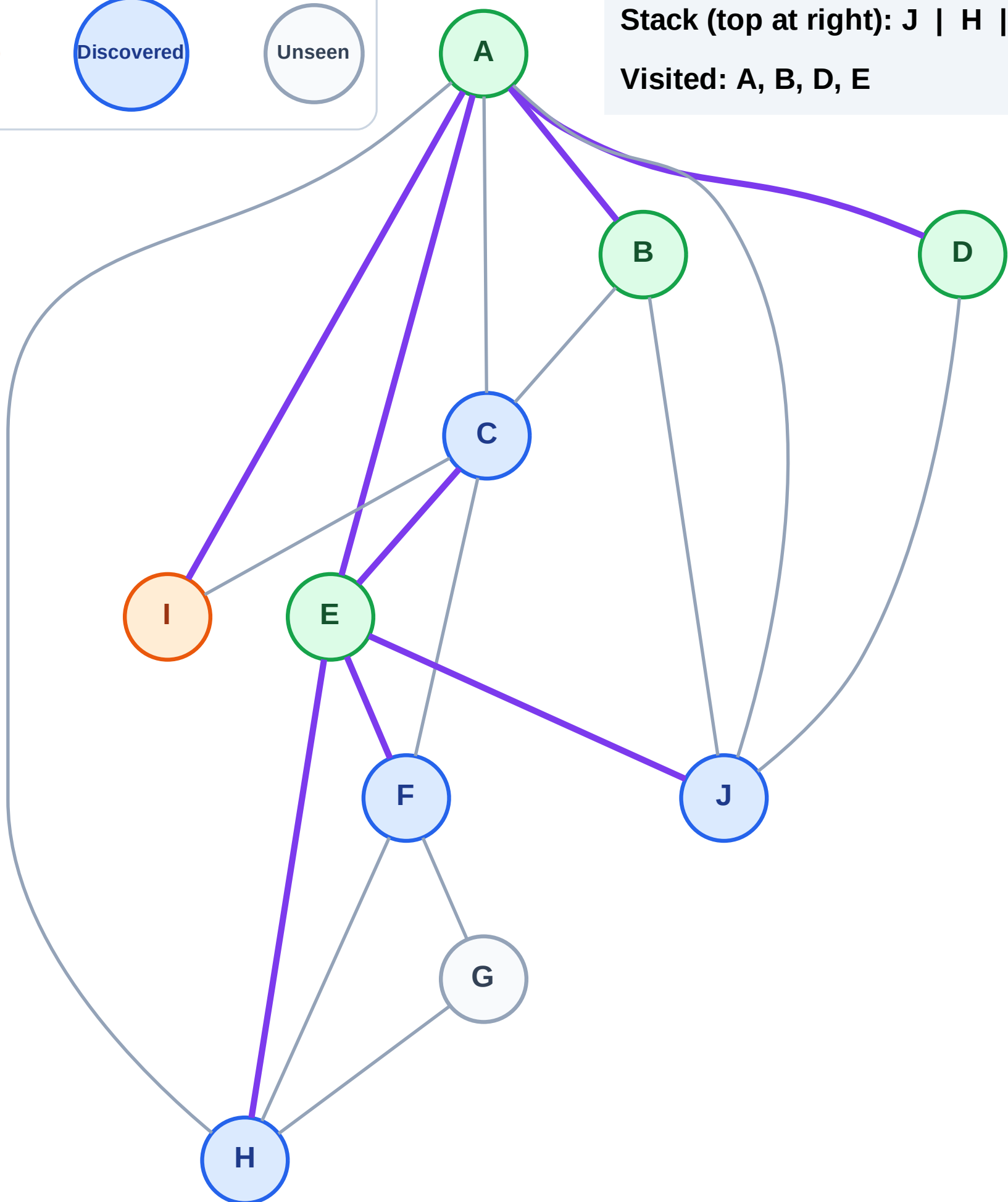
Processing

Discovered

Unseen

Stack (top at right): J | H | F | C

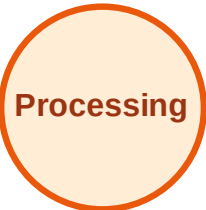
Visited: A, B, D, E



DFS Traversal

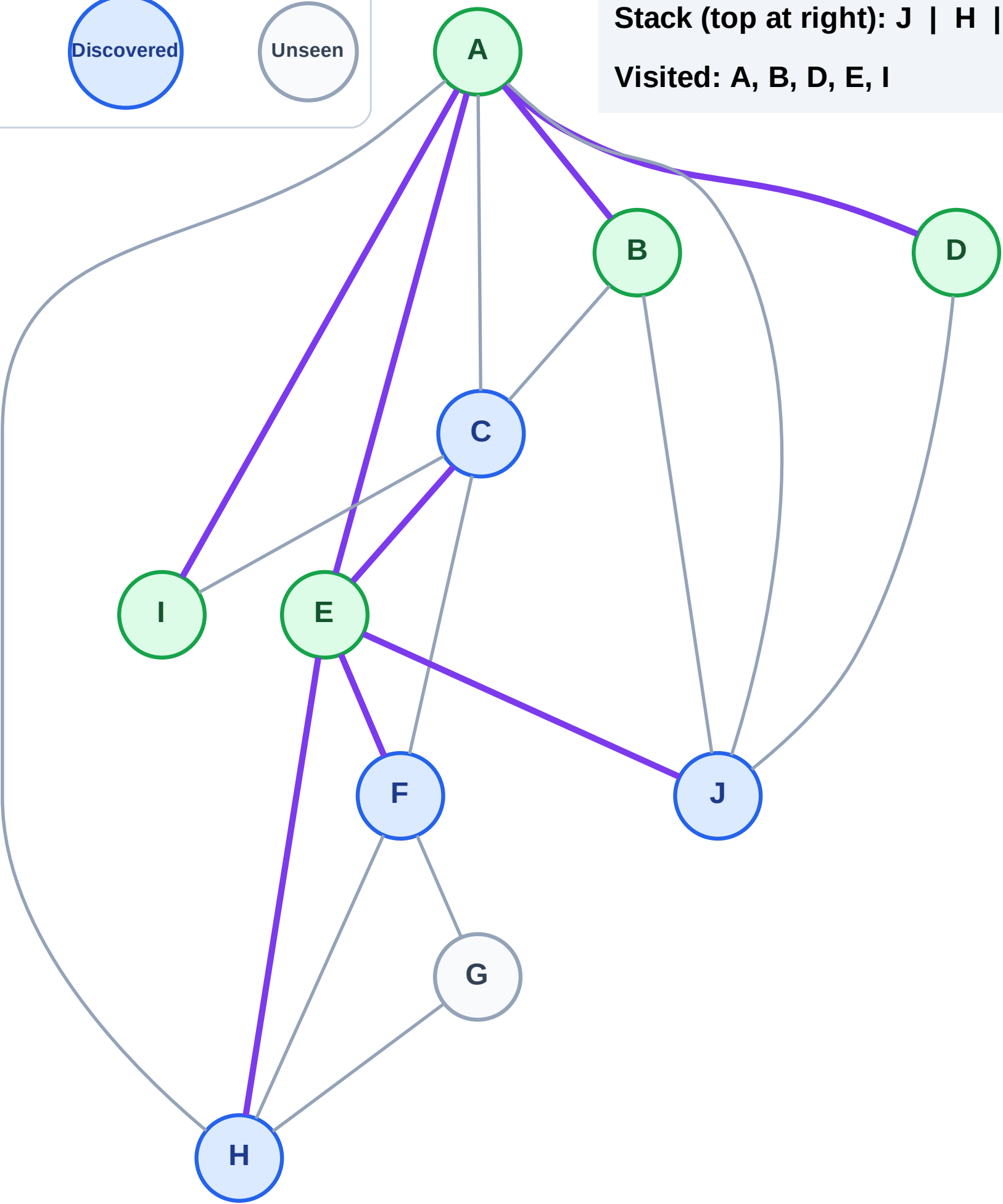
Step 11: No new neighbors from I

Legend



Stack (top at right): J | H | F | C

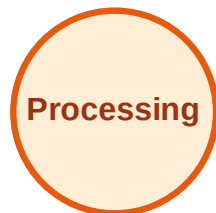
Visited: A, B, D, E, I



DFS Traversal

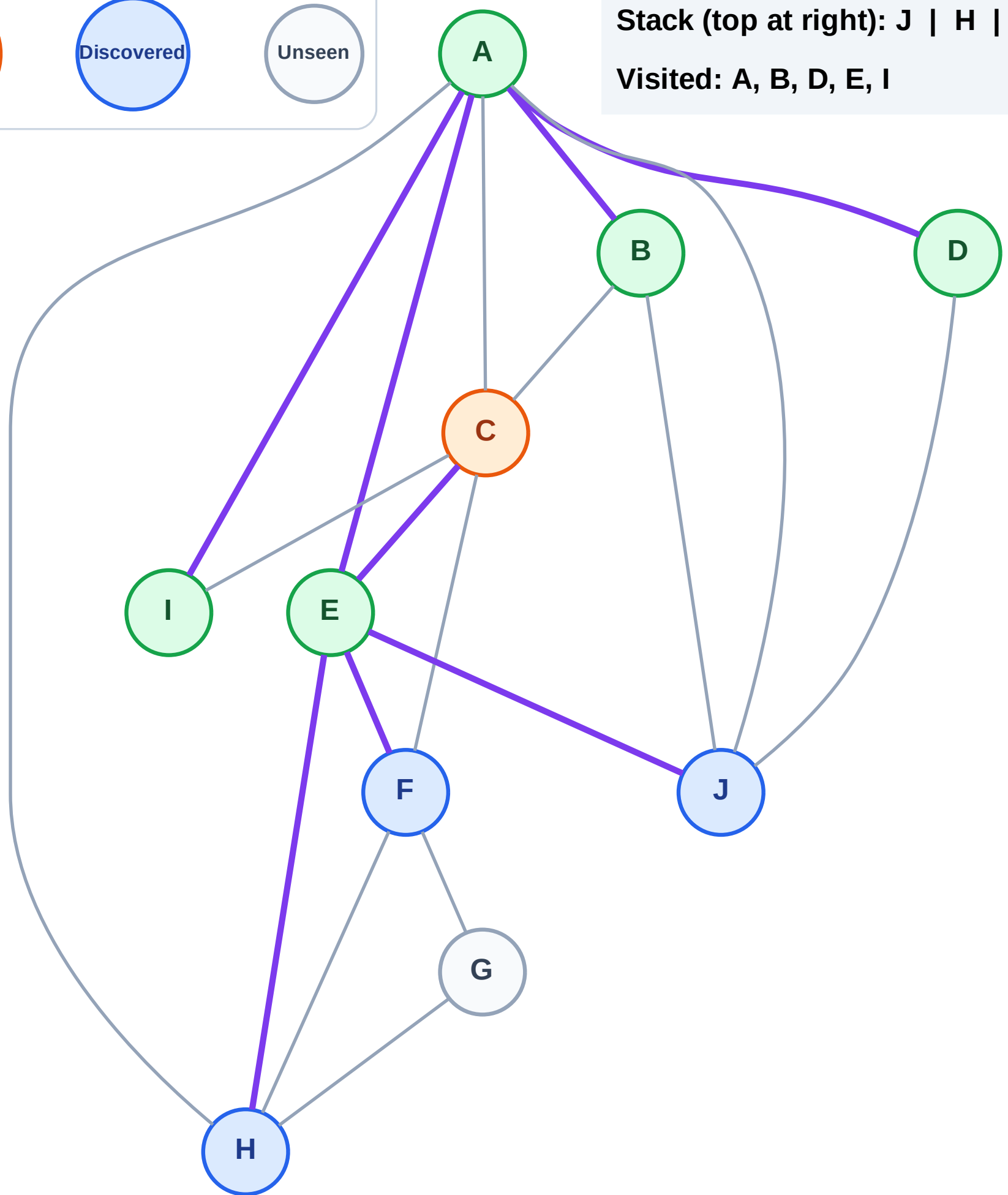
Step 12: Process node C

Legend



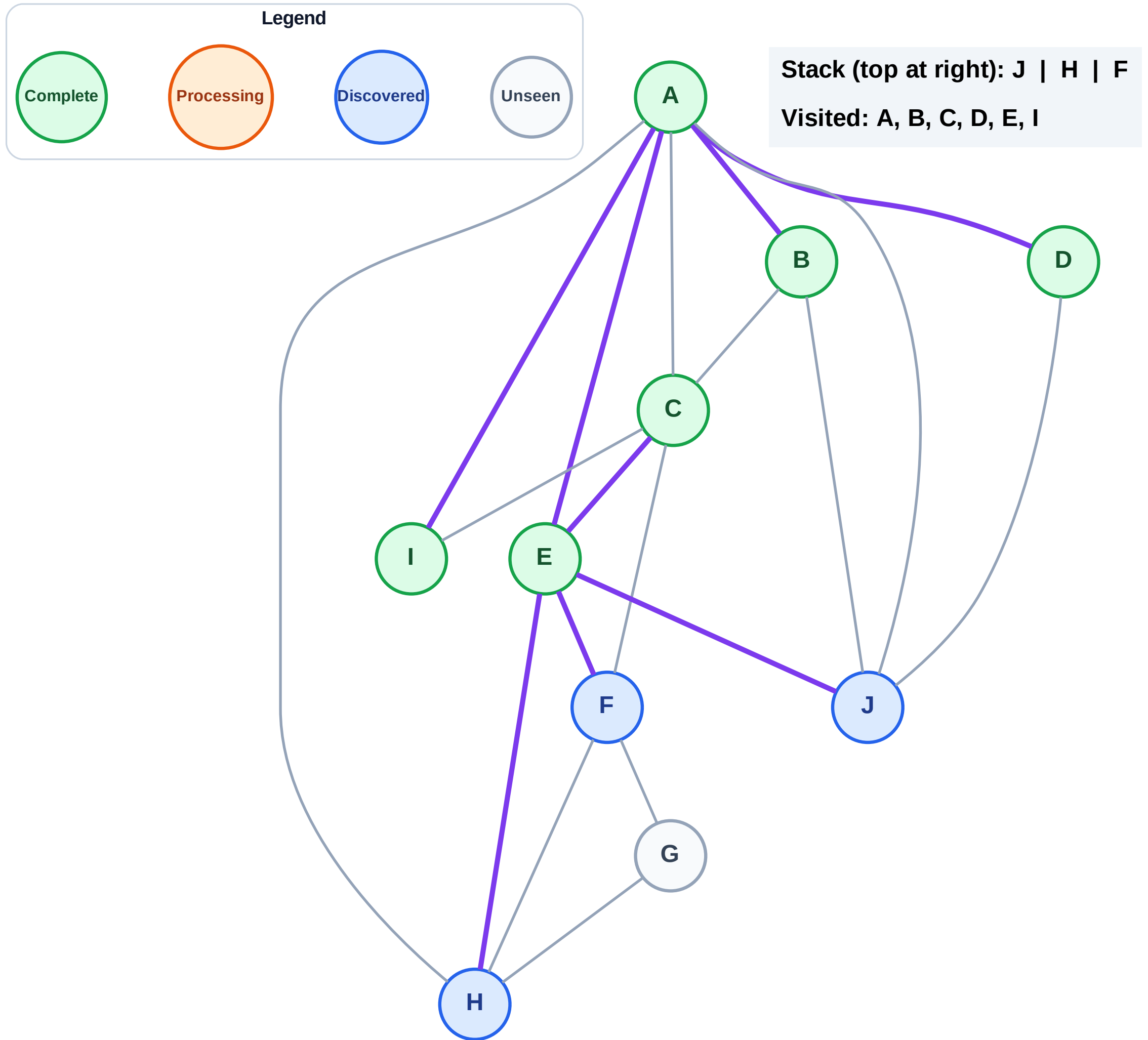
Stack (top at right): J | H | F

Visited: A, B, D, E, I



Step 13: No new neighbors from C

Step 13: No new neighbors from C



DFS Traversal

Step 14: Process node F

Legend

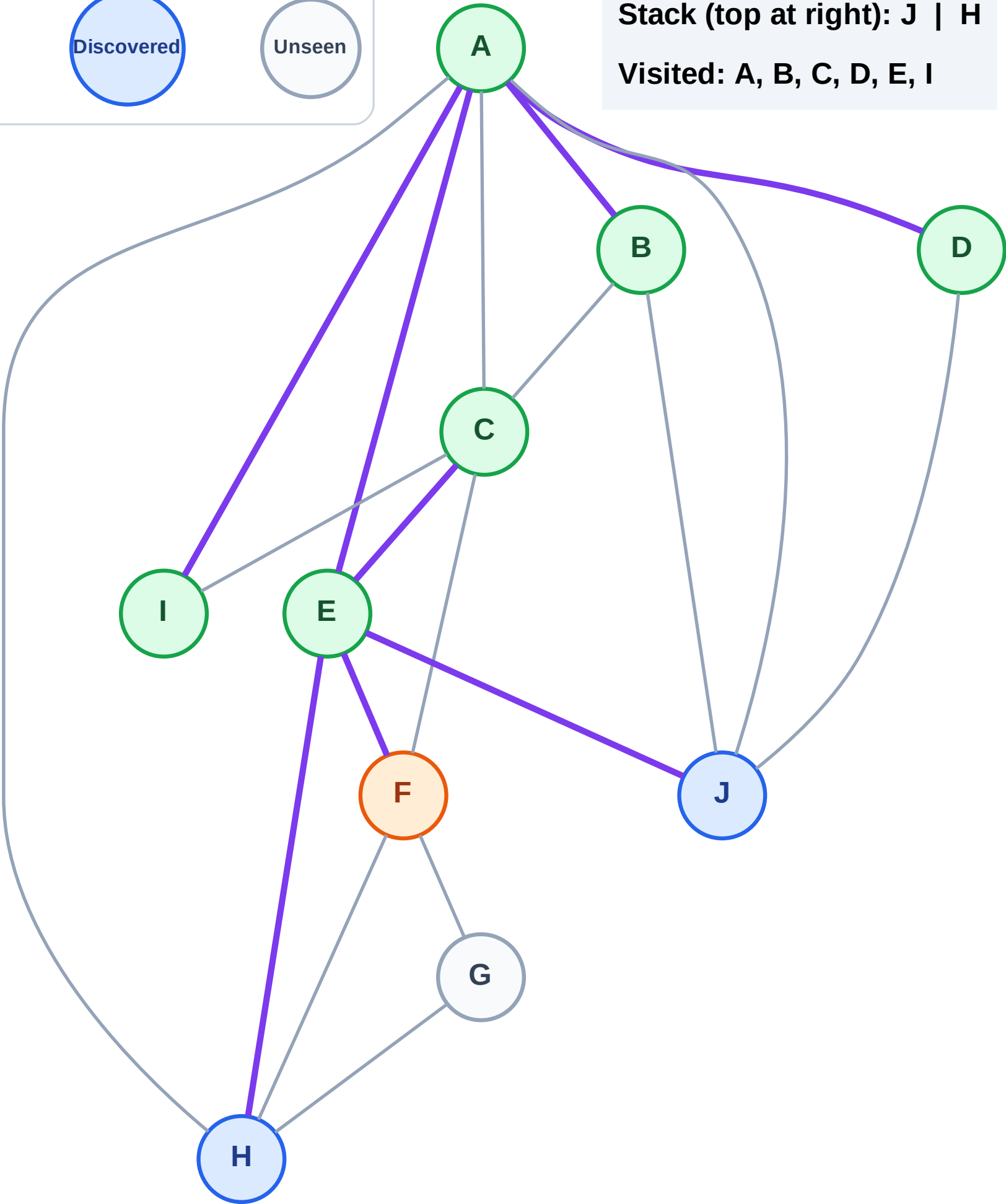
Complete

Processing

Discovered

Unseen

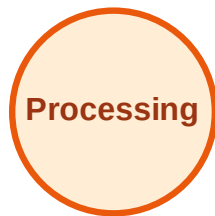
Stack (top at right): J | H
Visited: A, B, C, D, E, I



DFS Traversal

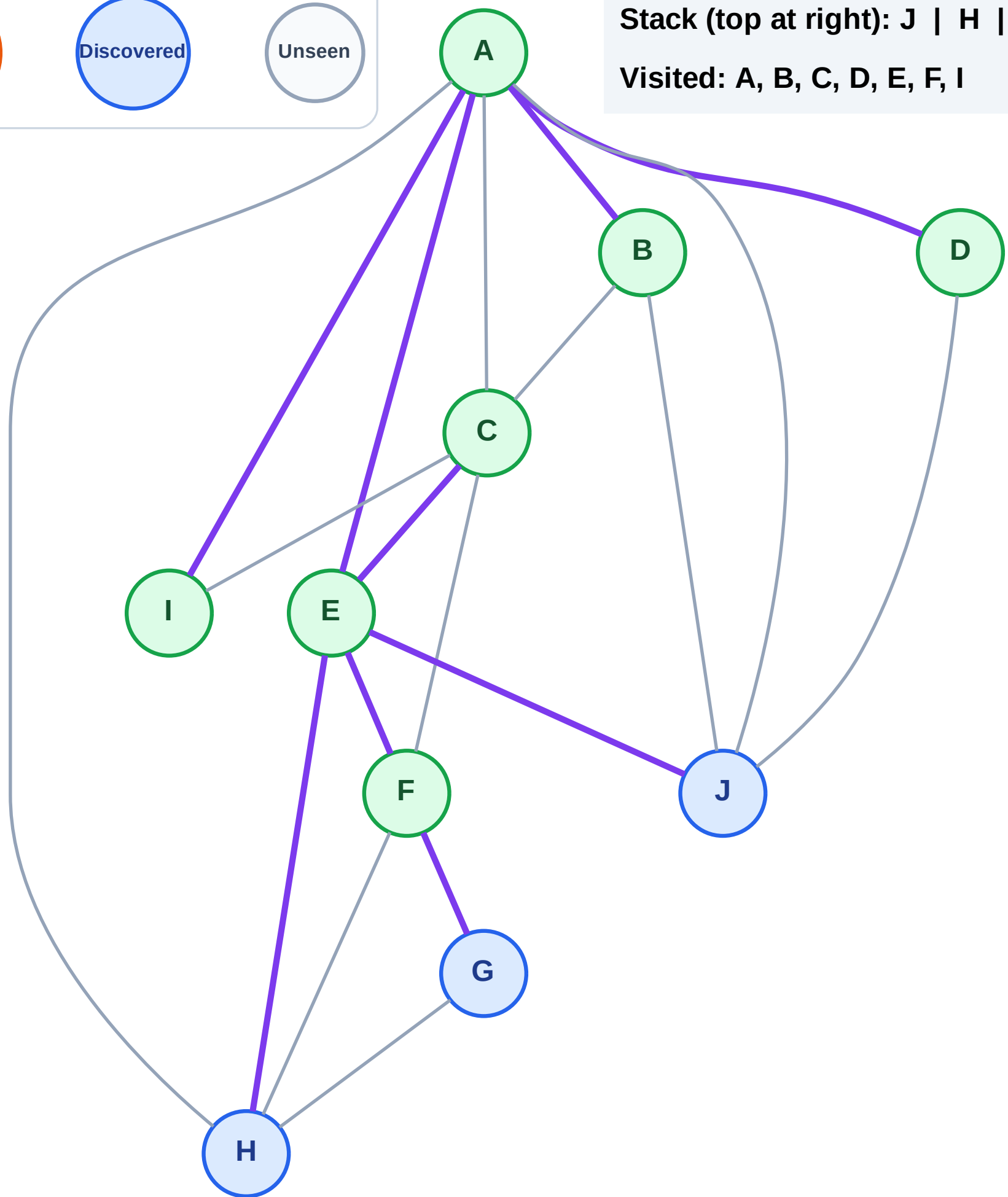
Step 15: Discover G from F

Legend



Stack (top at right): J | H | G

Visited: A, B, C, D, E, F, I



DFS Traversal

Step 16: Process node G

Legend

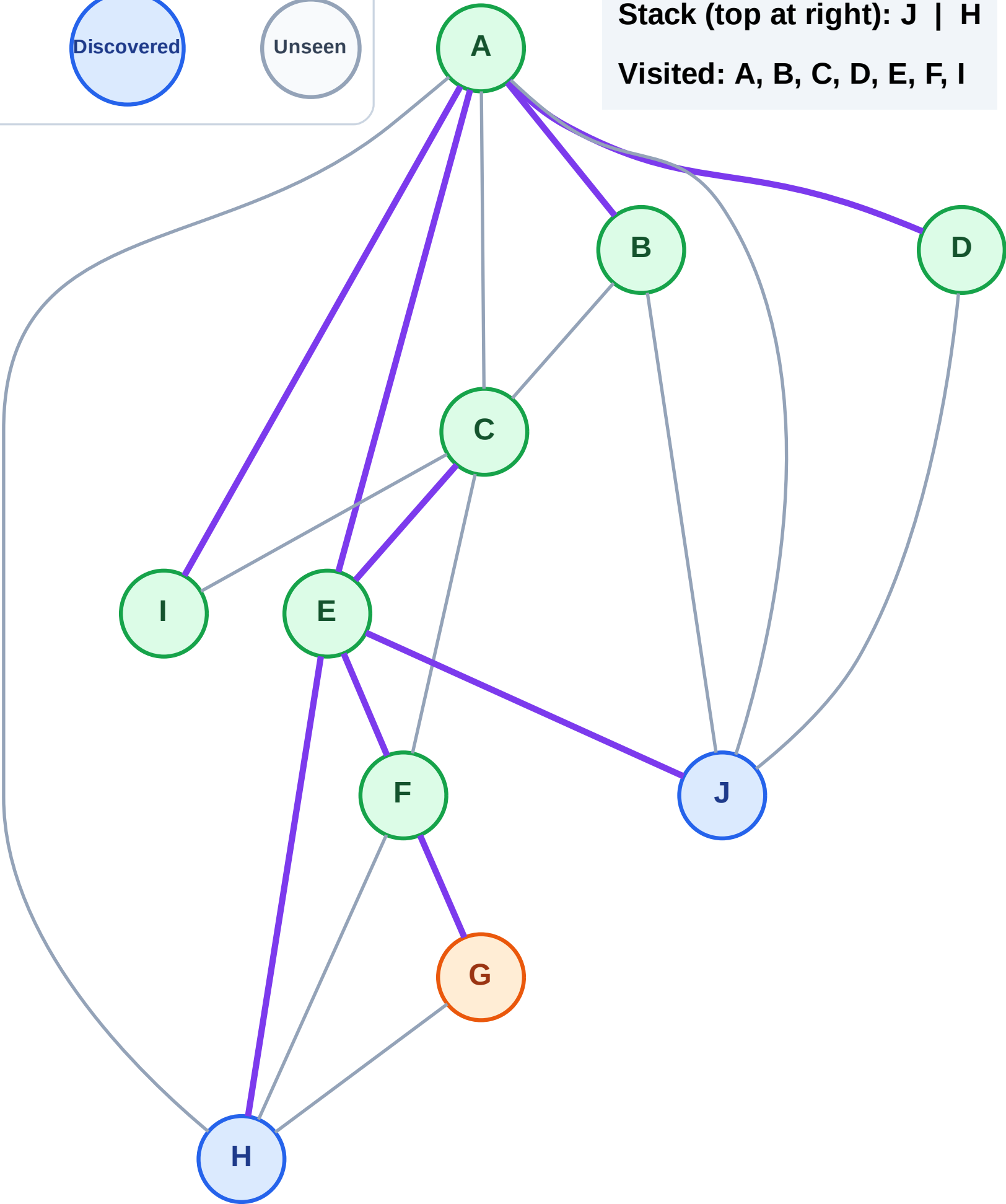
Complete

Processing

Discovered

Unseen

Stack (top at right): J | H
Visited: A, B, C, D, E, F, I



DFS Traversal

Step 17: No new neighbors from G

Legend

Complete

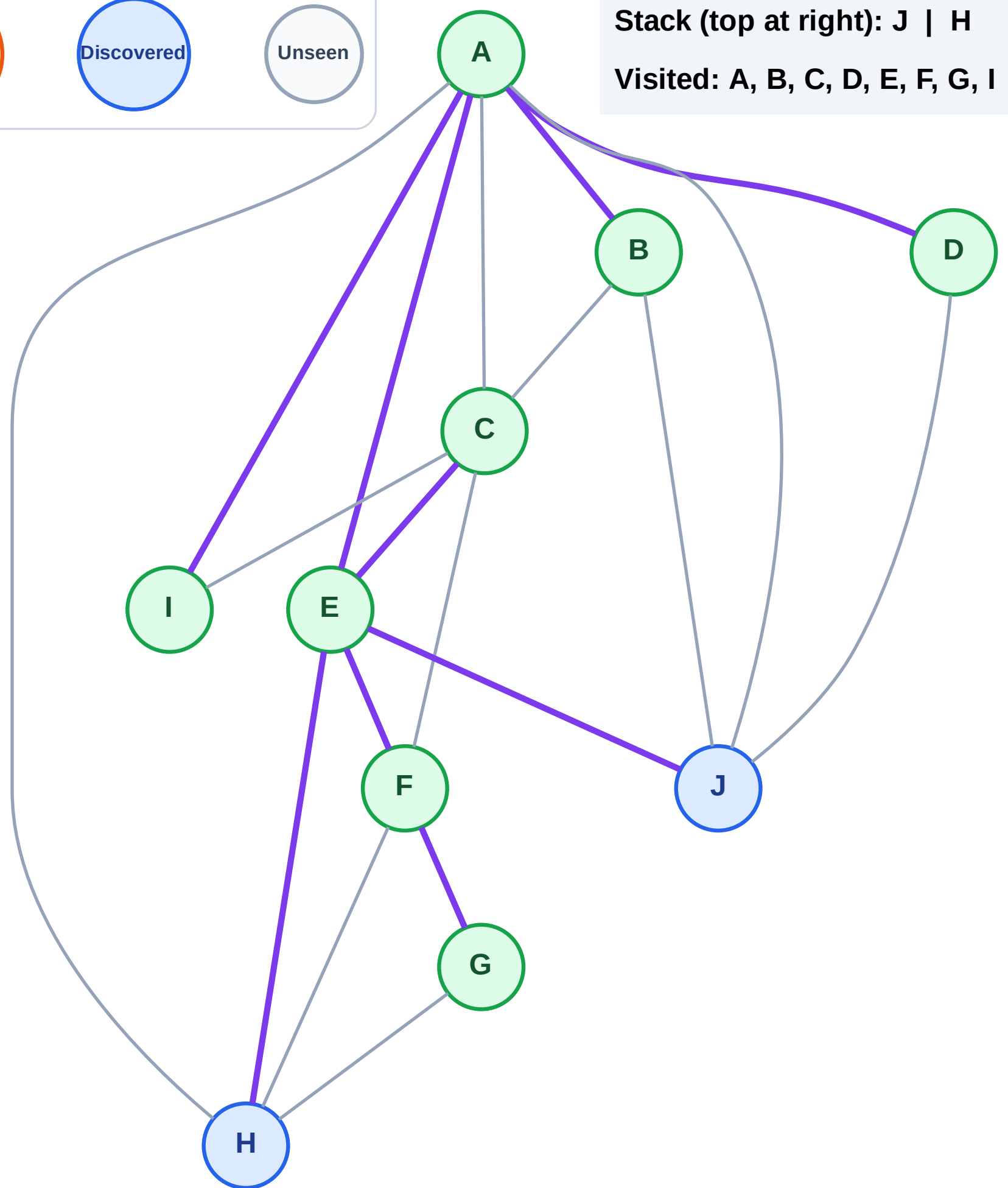
Processing

Discovered

Unseen

Stack (top at right): J | H

Visited: A, B, C, D, E, F, G, I



DFS Traversal

Step 18: Process node H

Legend

Complete

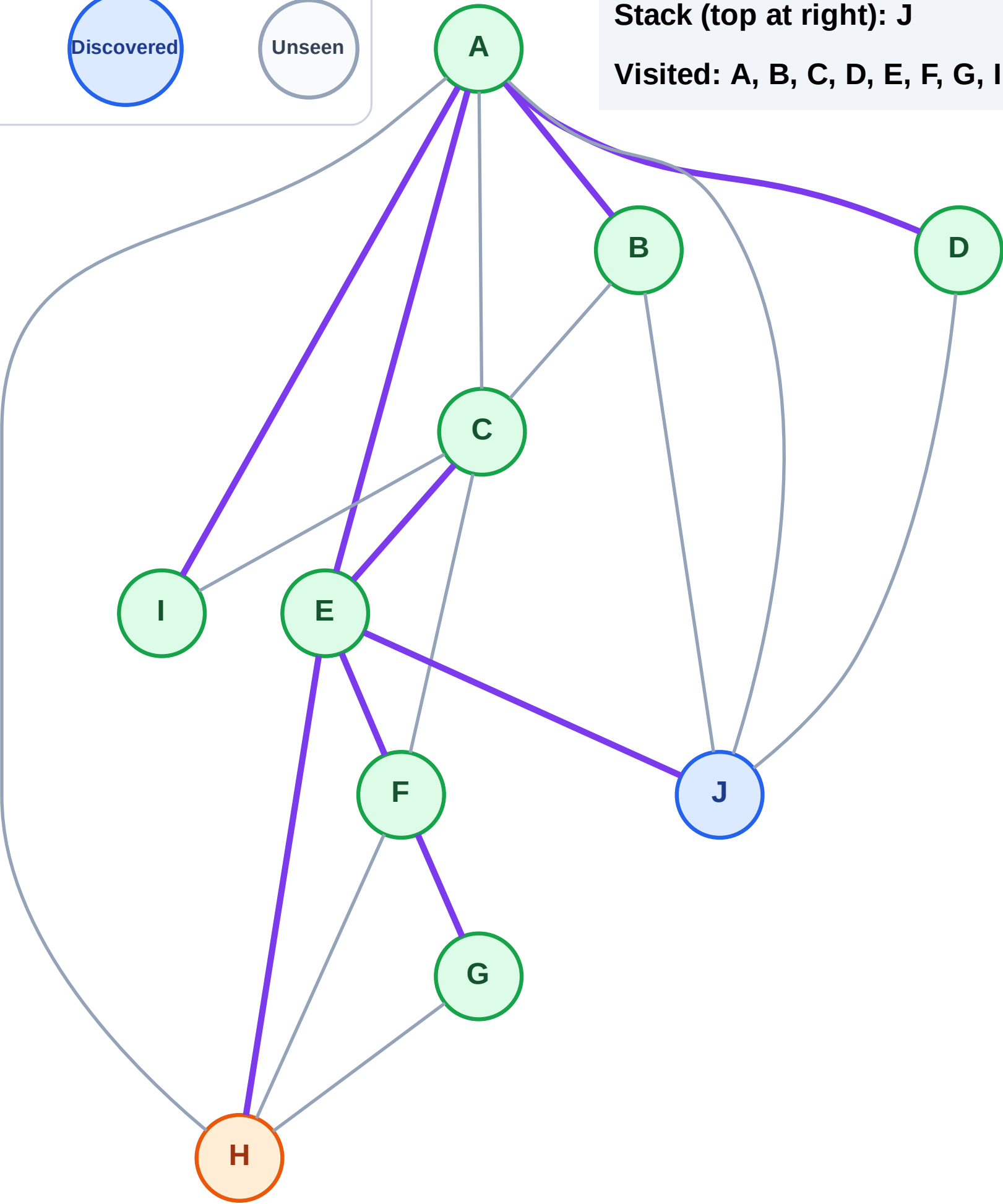
Processing

Discovered

Unseen

Stack (top at right): J

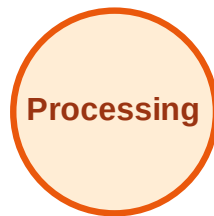
Visited: A, B, C, D, E, F, G, I



DFS Traversal

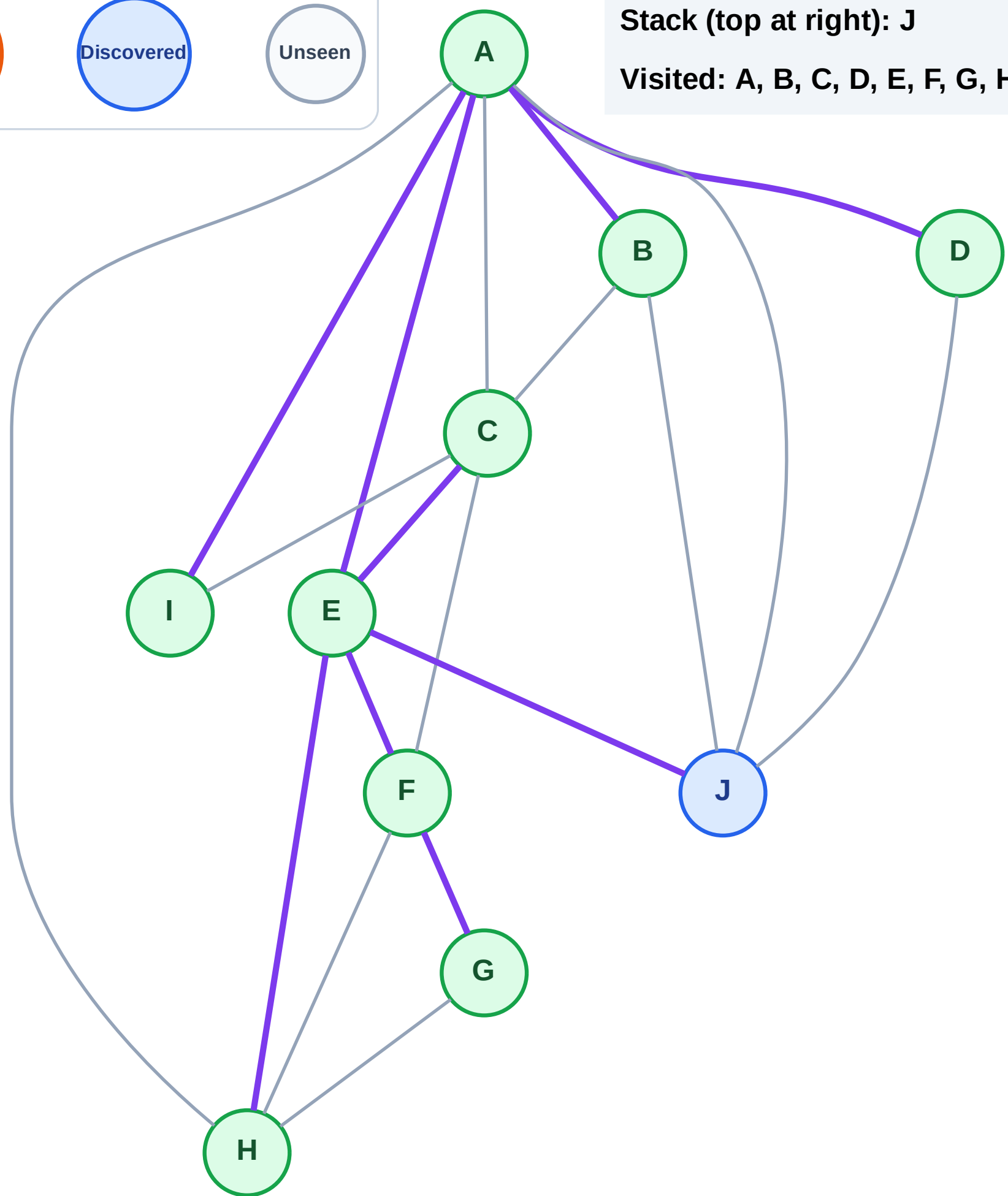
Step 19: No new neighbors from H

Legend



Stack (top at right): J

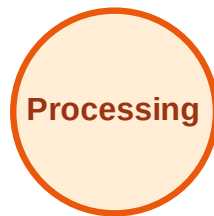
Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

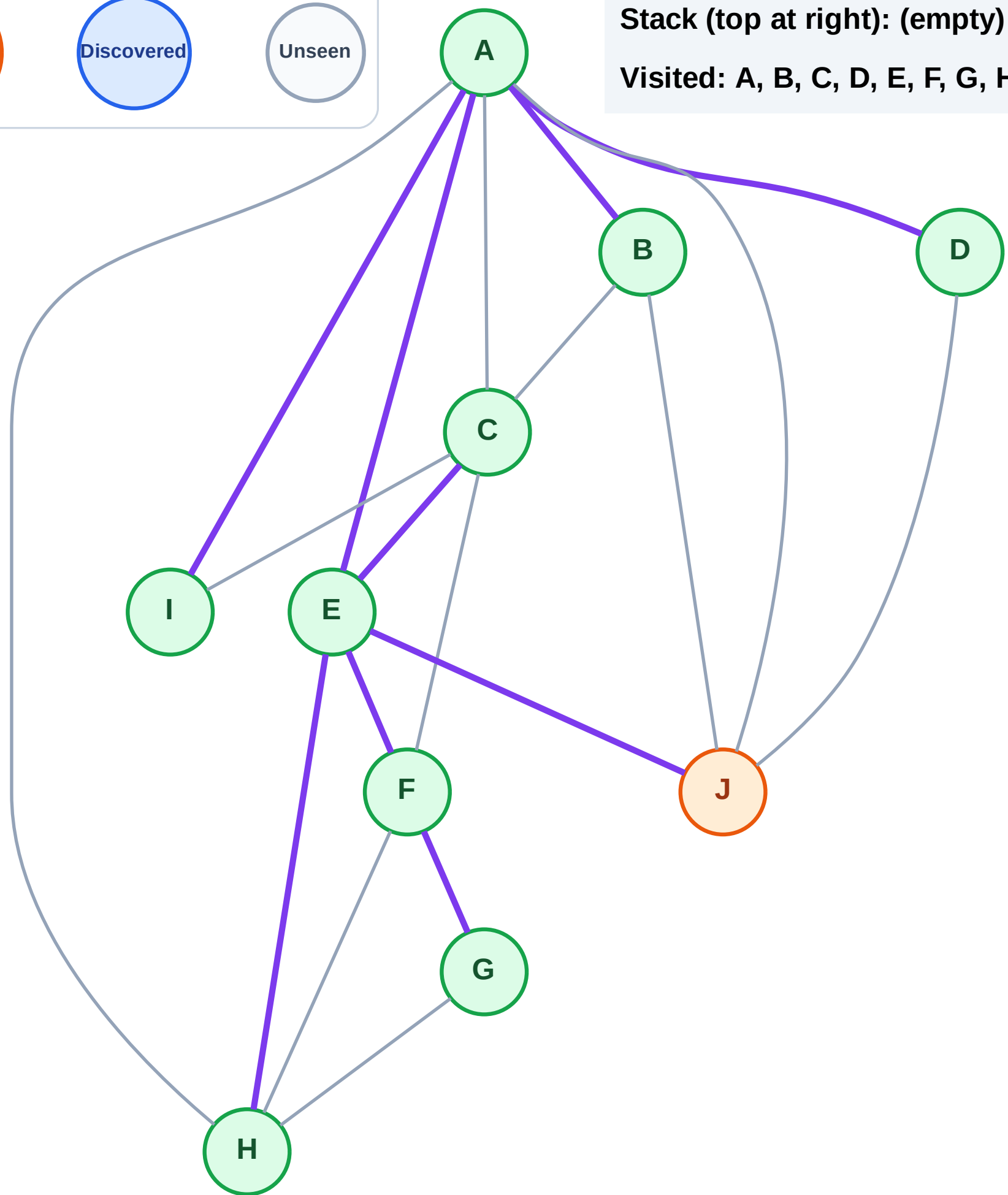
Step 20: Process node J

Legend



Stack (top at right): (empty)

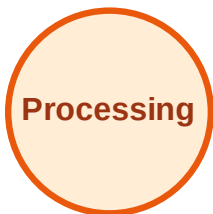
Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

Step 21: No new neighbors from J

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

